

Dong-A Univ. (ISPL)



동아대학교
DONG-A UNIVERSITY

Convolutional Neural Networks

컴퓨터공학부 AI학과
2024년 1학기 인공지능

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1. CNN 개요
2. Convolution 연산
3. 3D 데이터의 Convolution 연산



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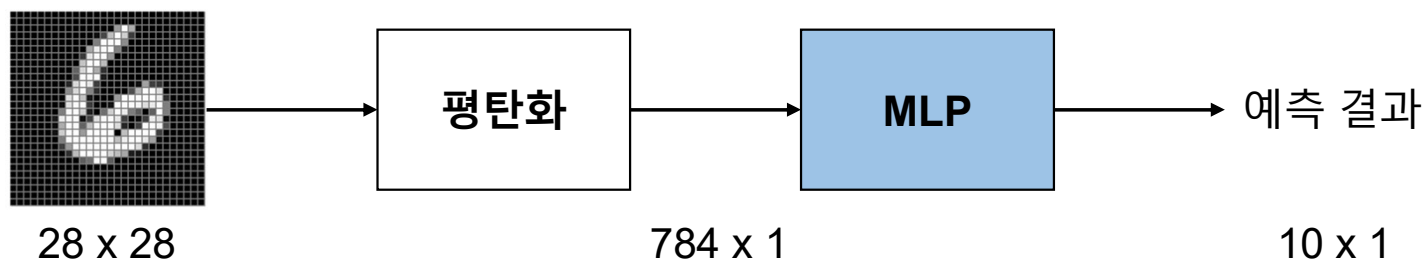
1. [CNN 개요](#)
2. Convolution 연산
3. 3D 데이터의 Convolution 연산



Convolutional Neural Networks - 개요

■ MLP 모델의 문제점

- 평탄화된 이미지를 입력으로 받아 공간적 (형상) 정보가 사라짐
- CNN에 비해 필요한 weight parameter의 개수가 많음

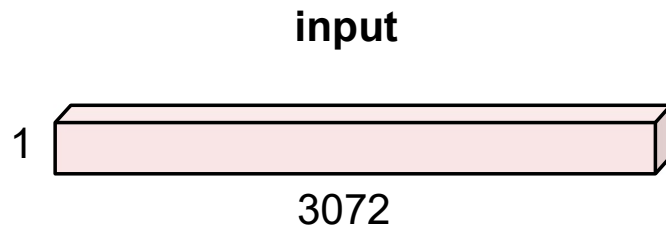


MLP의 입출력 구조 예시

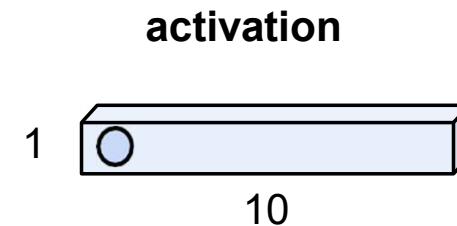
FC Layer의 문제점

- FC Layer는 1차원 데이터로 입력을 받음
- 따라서, 2D 이미지 데이터를 1D로 **평탄화** 해 입력함 → 이미지의 공간적(형상) 정보가 무시됨

ex) 32x32x3 image -> stretch to 3072 x 1



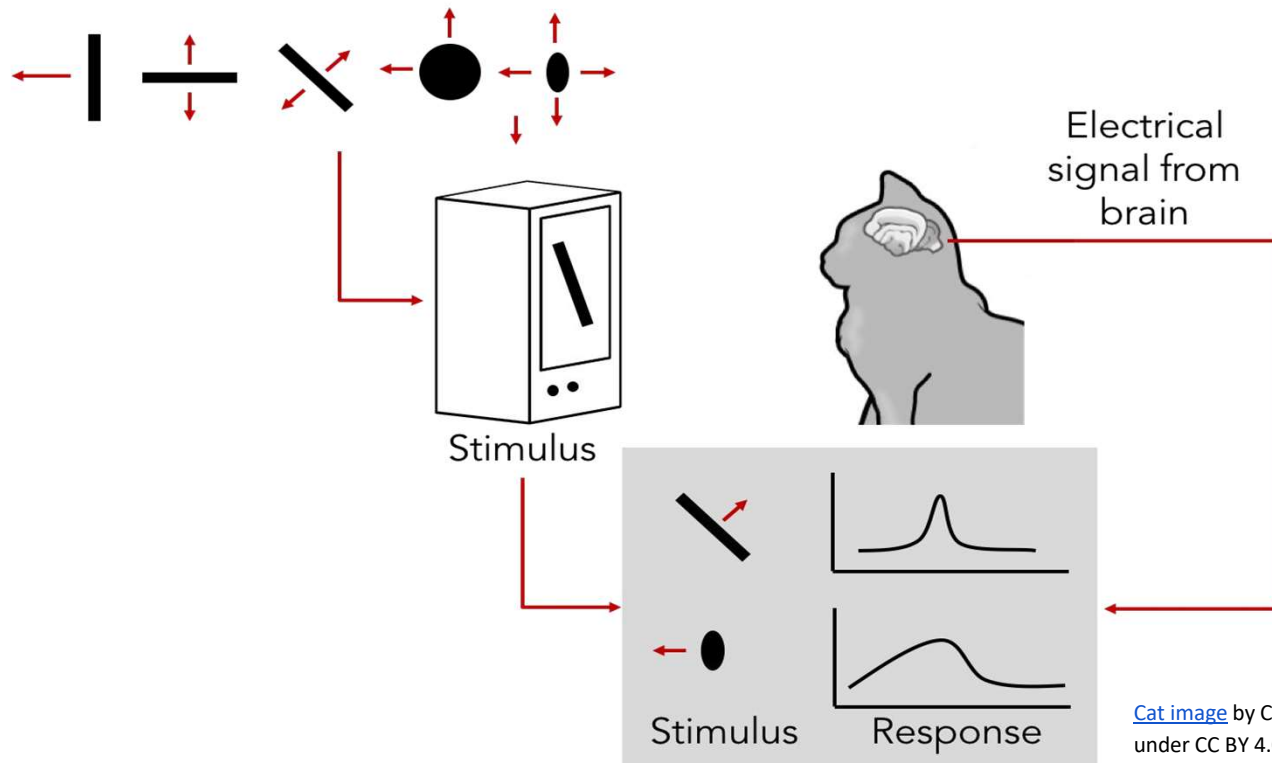
Wx
10 x 3072
weights



Fei-Fei Li & Justin Johnson & Serena Yeung

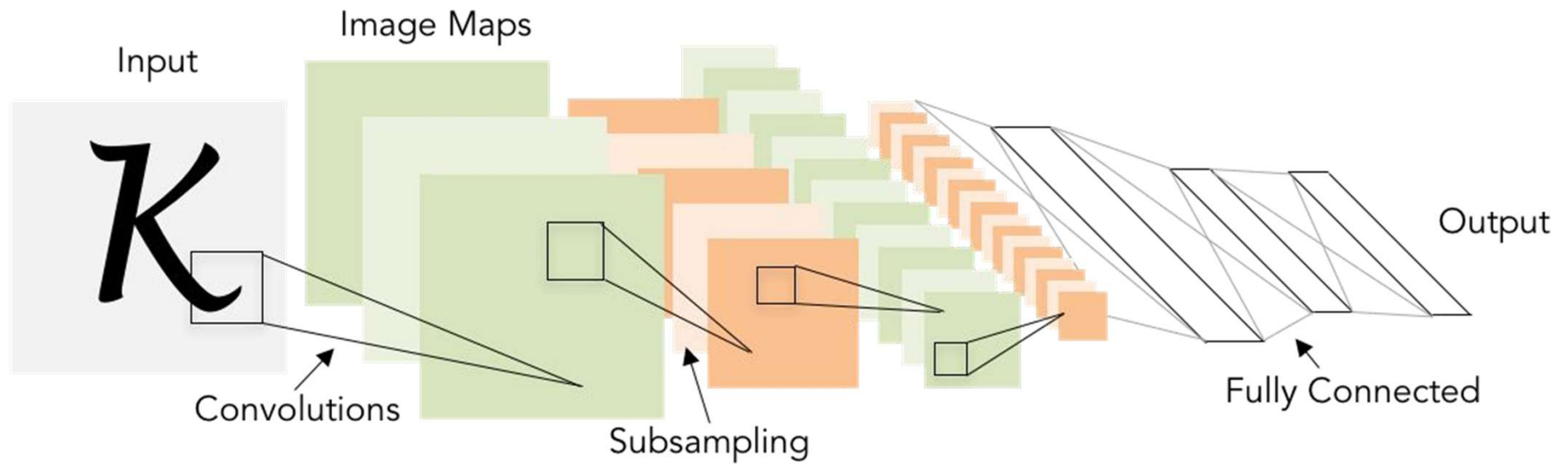
출처: cs231n_2017_lecture5 April 18, 2017

- 이미지의 특성에 따라 뉴런이 다르게 활성화 됨



[Cat image](#) by CNX OpenStax is licensed under CC BY 4.0; changes made

- 2D이미지를 Input으로 사용



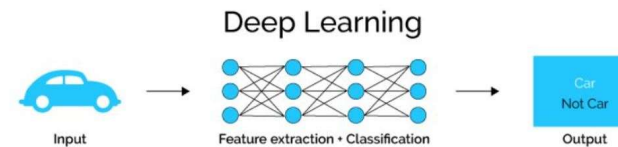
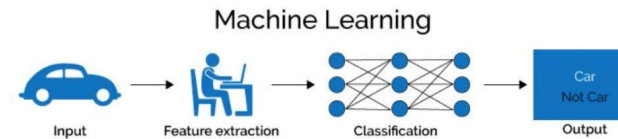
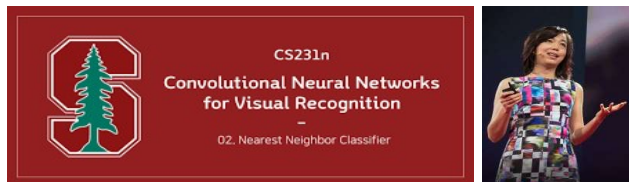
ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

What is ImageNet ?



ImageNet Visual Dataset (Source)

ImageNet is a visual Dataset that contains more than **15 million of labeled high-resolution** images covering almost **22,000 categories**. ImageNet Dataset is of high quality and that's one of the reasons it is highly popular among researchers to test their image classification model on this dataset.



‘딥러닝 혁명’의 아버지들 컴퓨터 과학의 노벨상 튜링상 수상

2019.03.28 19:47

| AI의 기초기술이 되는 콘볼루션 신경망 개발



튜링상 수상자로 선정된 조프리 힌튼, 조슈아 벤지오, 안 르쿤. 연합뉴스/AP 제공

Google

facebook

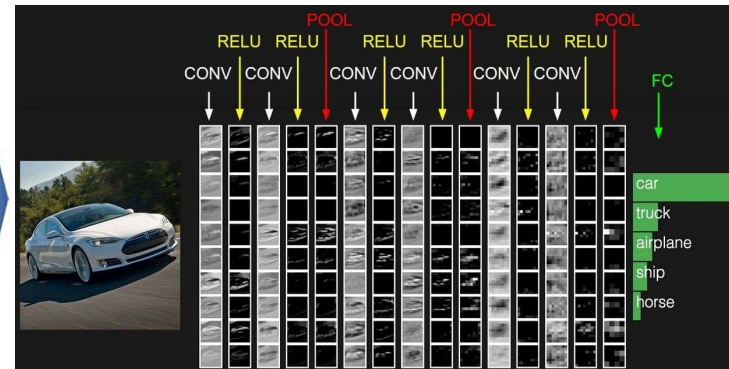
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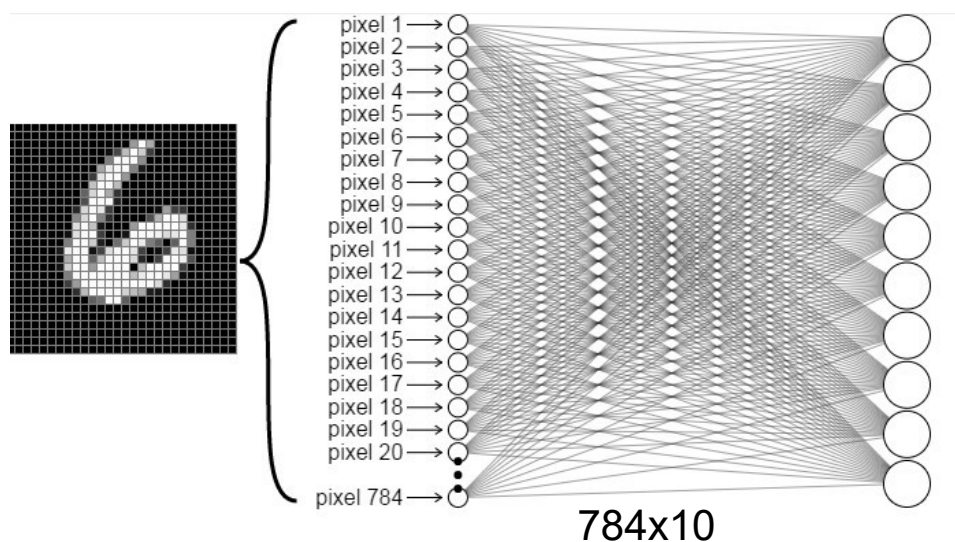
우승 알고리즘의 분류 에러율(%)



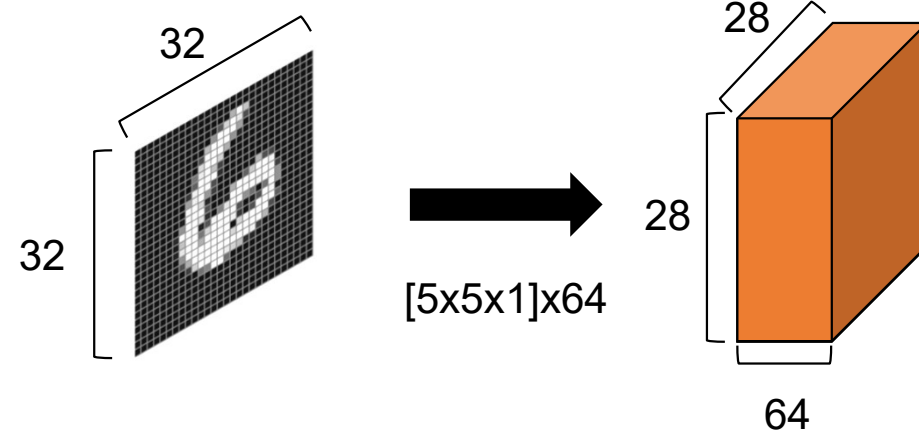
Convolutional Neural Networks - 개요

■ Perceptron 모델의 문제점

- 평탄화된 이미지를 입력으로 받아 공간적 (형상) 정보가 사라짐
- CNN에 비해 필요한 weight parameter의 개수가 많음



Fully connected layer
(#weights: 7,840)

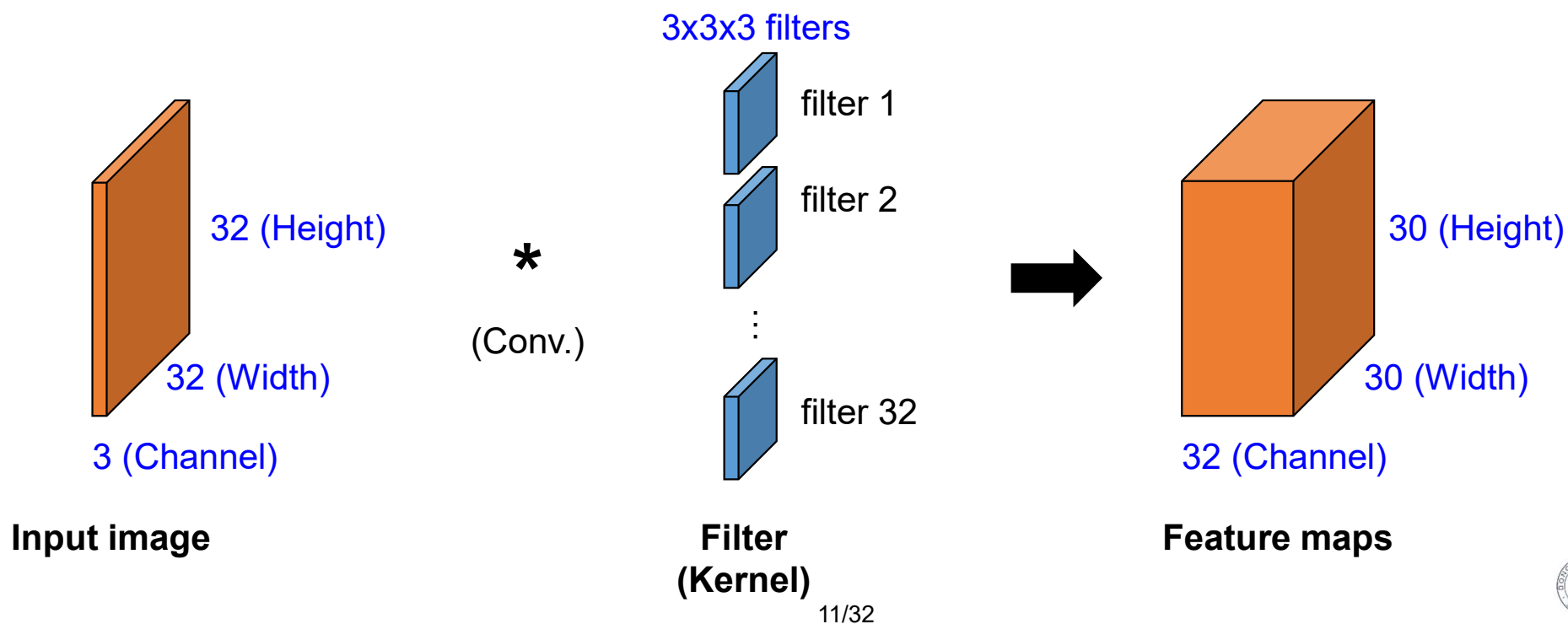


5x5 Convolution layer
(#weights: 1,600)

Convolutional Neural Networks - 개요

■ CNN 모델 개요

- 합성곱 연산 (Convolution)을 수행해 이미지의 특징 (Feature) 추출
- 이미지를 입력으로 받는 분야에서 높은 성능을 보임 (Ex. Image classification, Super-resolution, Denoising)



▪ Democratization of AI

[AI 민주화] 누구나 AI로부터 차별 받지 않고 AI 기술에 쉽게 접근하고 개발할 수 있도록 하는 것!



과기정통부와 한국지능정보사회진흥원(NIA)은 인공지능(AI) 학습용 데이터 170종을 "AI 허브"에 단계적으로 개방한다고 밝혔다. (2021년 6월 18일)



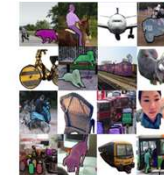
<CIFAR-10>



<Caltech 101>



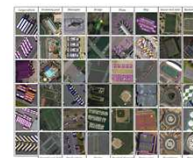
<MNIST>



<MS-COCO>



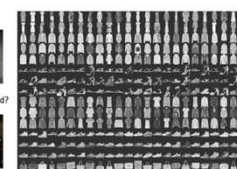
<Columbia University Image Library>



<DOTA>



<Visual QA>



<Fashion-MNIST>

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Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

4x4x1 image

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

3x3x1 filter

2	0	1
0	1	2
1	0	2

Filter
(Kernel)

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

입력 이미지와 필터의 채널은 같아야 함

4x4x1 image

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

3x3x1 filter

2	0	1
0	1	2
1	0	2

Filter
(Kernel)

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

*

Conv.
연산

2	0	1
0	1	2
1	0	2

Filter (Kernel)

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

$$\text{Output} = 1 \times 2 + 2 \times 0 + 3 \times 1 + 0 \times 0 + 1 \times 1 + 2 \times 2 + 3 \times 1 + 0 \times 0 + 1 \times 2$$

$$= 15$$

15	

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

*

Conv.
연산

2	0	1
0	1	2
1	0	2

Filter (Kernel)

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

*
Conv.
연산

2	0	1
0	1	2
1	0	2

Filter (Kernel)

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

15	

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

15	16

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

15	16
6	

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

15	16
6	15

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

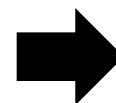
1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

$*$
Conv.
연산

2	0	1
0	1	2
1	0	2

Filter (Kernel)

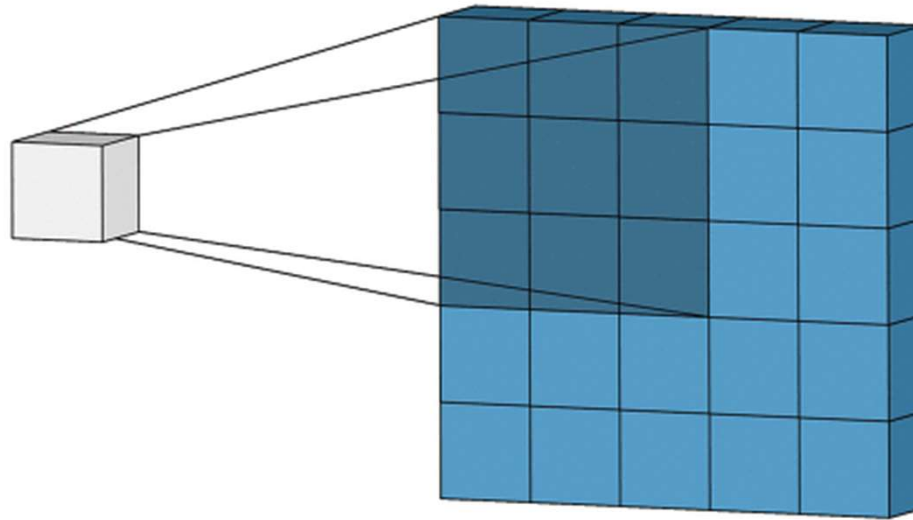


15	16
6	15

Output feature map

Convolutional Neural Networks - Convolution 연산

- 2D 이미지 입력에 대한 convolution 연산 예시

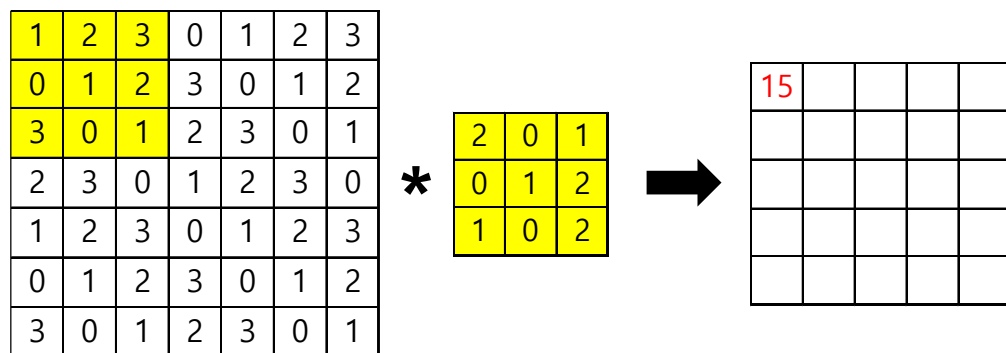


Output feature map ← Input image

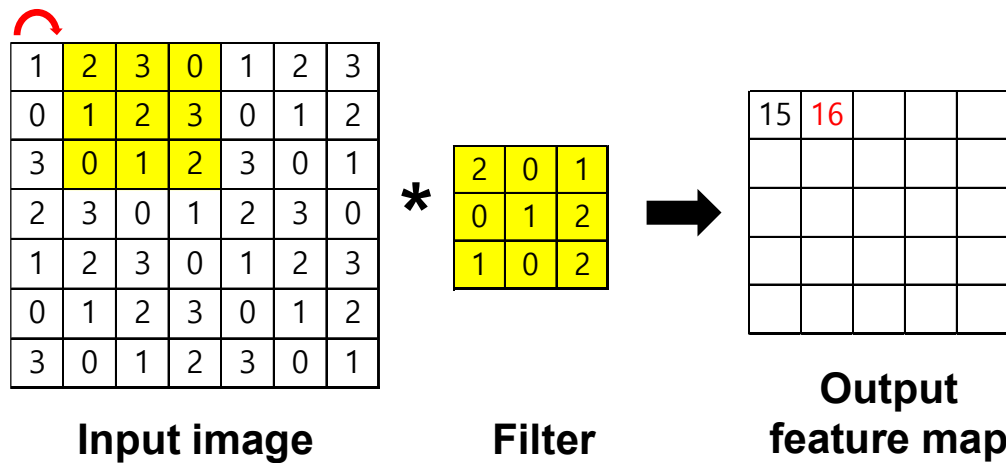
Convolutional Neural Networks - Convolution 연산

▪ Stride: Convolution 연산의 step size

- Stride가 커질수록 feature map의 크기는 작아짐



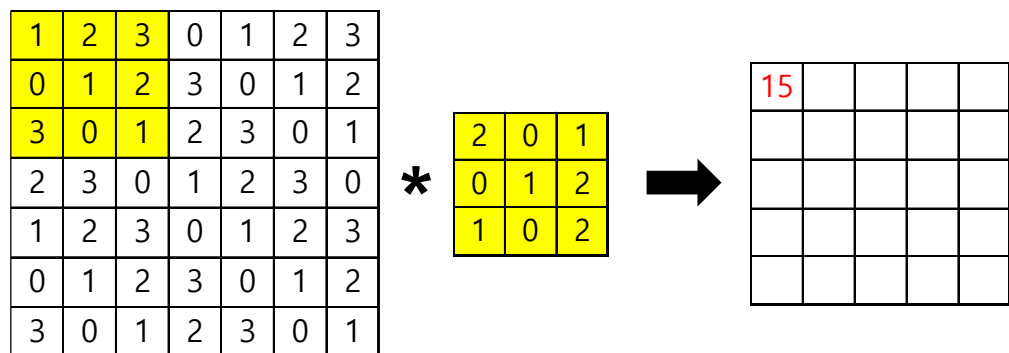
Stride=1



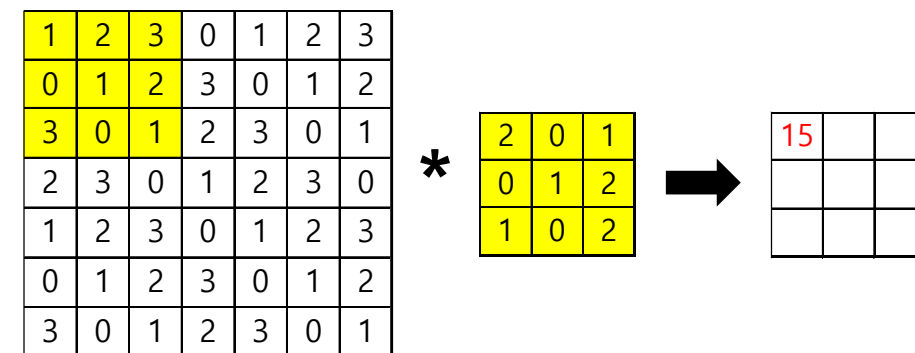
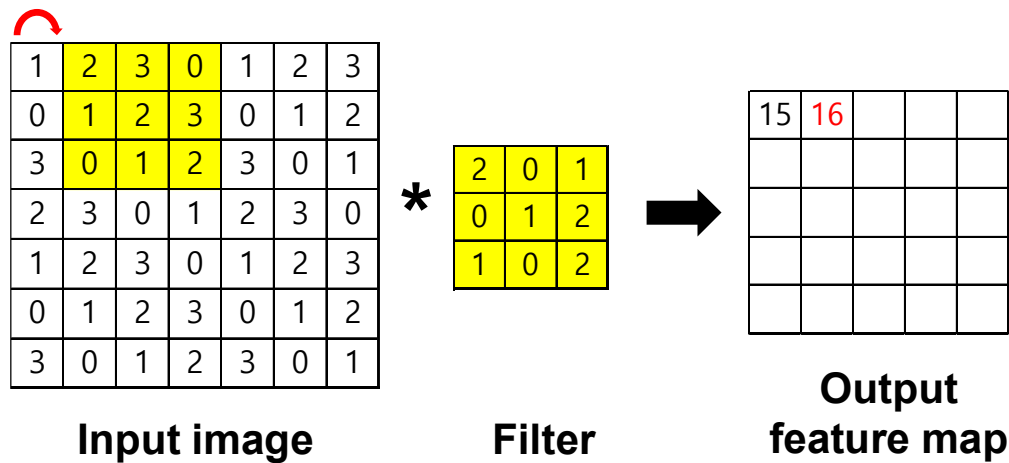
Convolutional Neural Networks - Convolution 연산

Stride: Convolution 연산의 step size

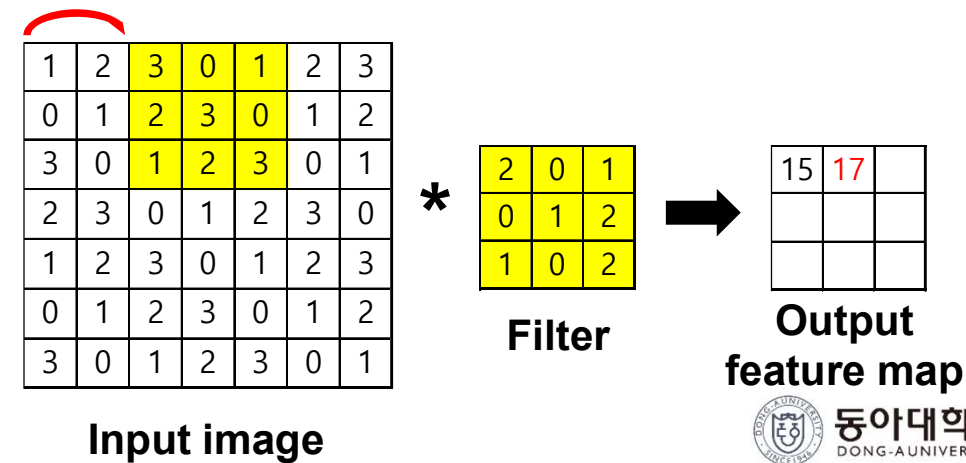
- Stride가 커질수록 feature map의 크기는 작아짐



Stride=1



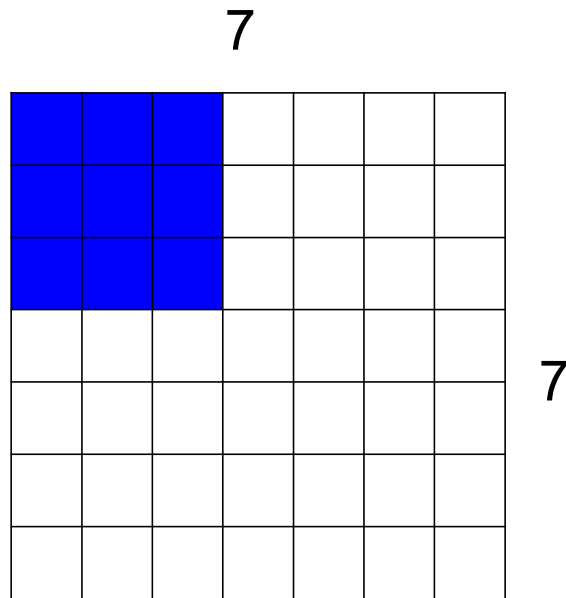
Stride=2



Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

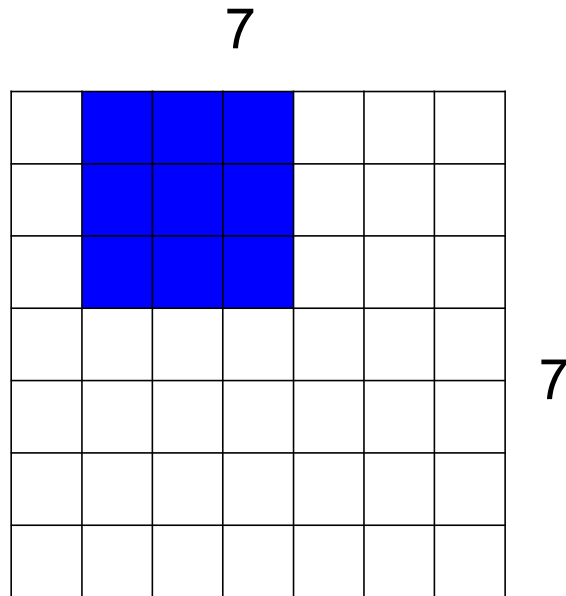


7x7 input (spatially)
assume 3x3 filter

Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

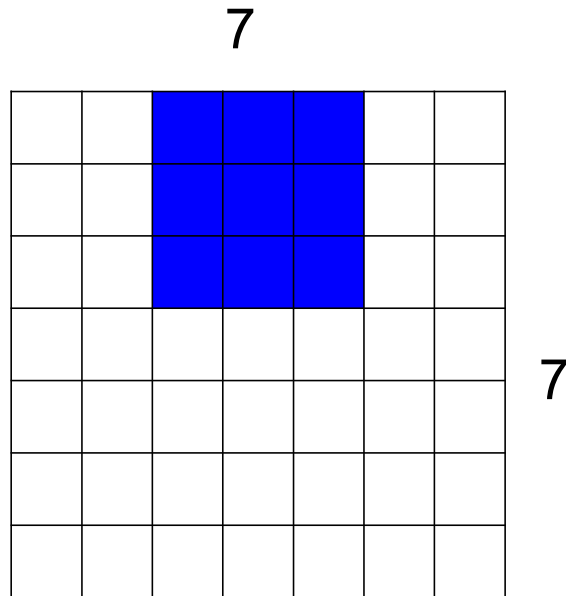


7x7 input (spatially)
assume 3x3 filter

Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

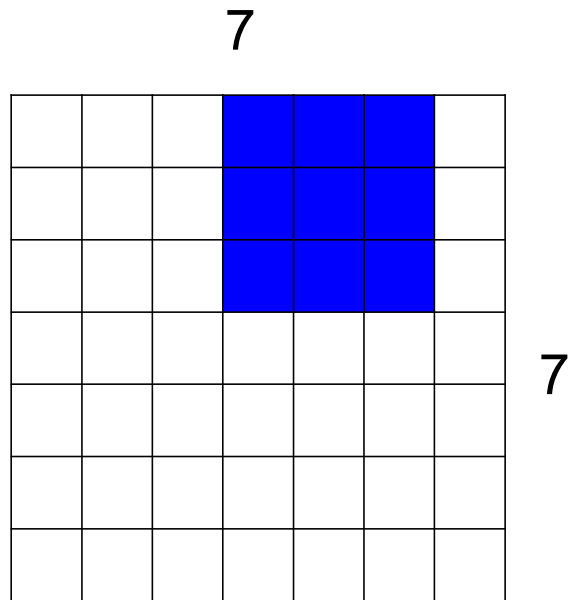


7x7 input (spatially)
assume 3x3 filter

Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

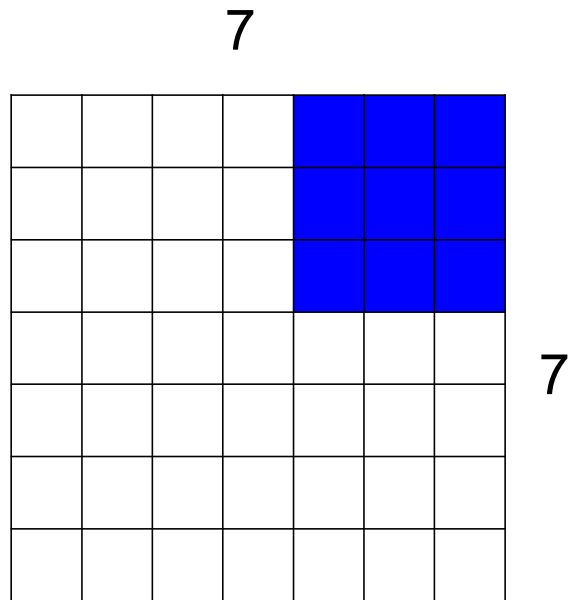


7x7 input (spatially)
assume 3x3 filter

Convolution Layer

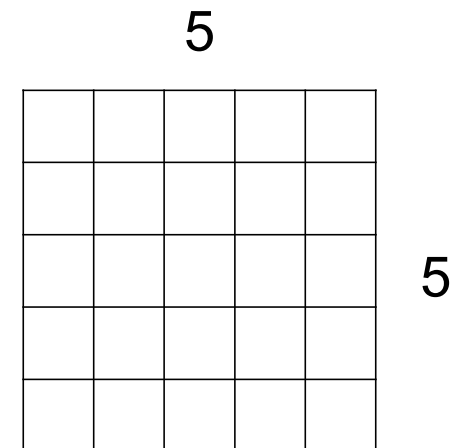
▪ Stride

- Filter가 Convolution 연산을 수행하는 Step size



7x7 input (spatially)
assume 3x3 filter

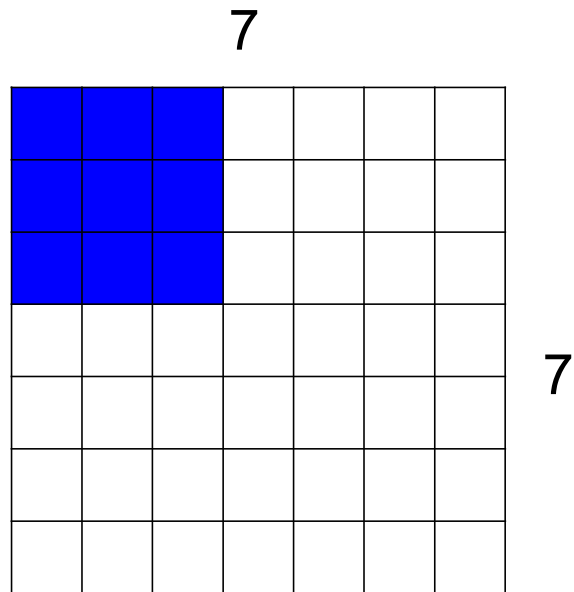
=> 5x5 output



Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

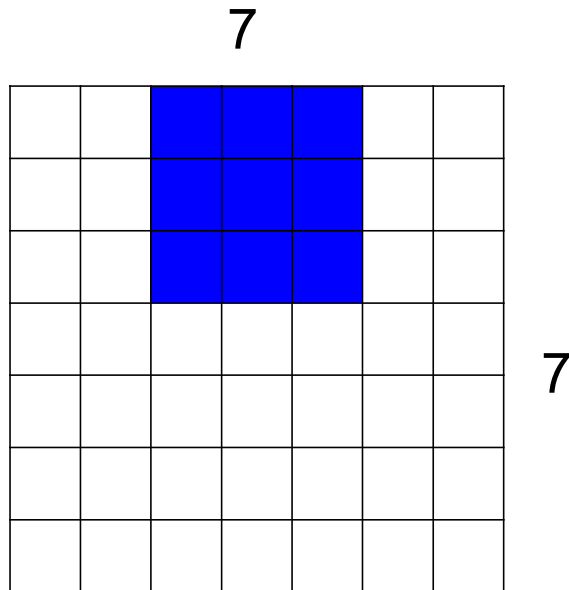


7x7 input (spatially)
assume 3x3 filter applied **with stride 2**

Convolution Layer

- **Stride**

- Filter가 Convolution 연산을 수행하는 Step size

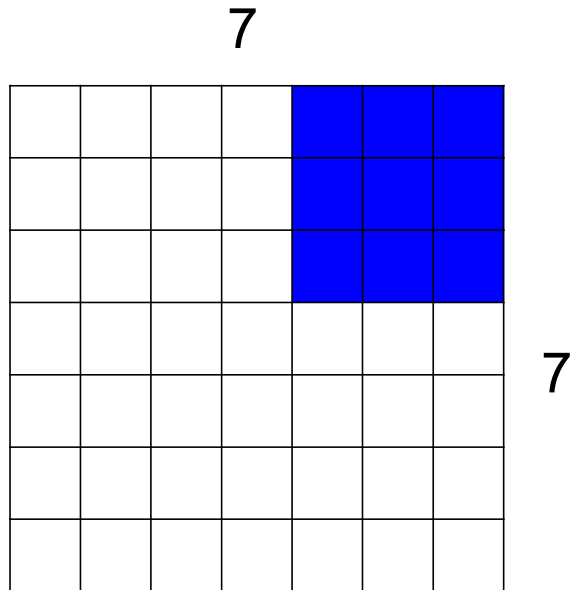


7x7 input (spatially)
assume 3x3 filter applied **with stride 2**

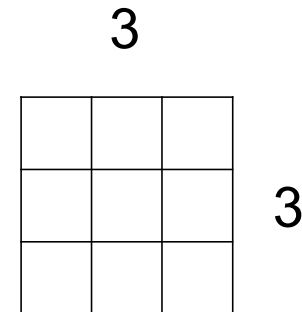
Convolution Layer

▪ Stride

- Filter가 Convolution 연산을 수행하는 Step size



7x7 input (spatially)
assume 3x3 filter applied **with stride 2**
=> 3x3 output!



Convolutional Neural Networks - Convolution 연산

- **Padding:** Input image 주변 값을 특정 값 (주로 0) 으로 채워 줌
 - Convolution 연산으로 boundary 정보가 소실되는 문제를 방지

Padding size = 0

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

stride: 1

2	0	1
0	1	2
1	0	2

Filter



15	16
6	15

**Output
feature map**

Convolutional Neural Networks - Convolution 연산

- Padding: Input image 주변 값을 특정 값 (주로 0) 으로 채워 줌
 - Convolution 연산으로 boundary 정보가 소실되는 문제를 방지

Padding size = 0

1	2	3	0
0	1	2	3
3	0	1	2
2	3	0	1

Input image

*
stride: 1

2	0	1
0	1	2
1	0	2

Filter



15	16
6	15

Output
feature map

Padding size = 1

0	0	0	0	0	0
0	1	2	3	0	0
0	0	1	2	3	0
0	3	0	1	2	0
0	2	3	0	1	0
0	0	0	0	0	0

Input image

*
stride: 1

2	0	1
0	1	2
1	0	2

Filter



7	12	10	2
4	15	16	10
10	6	15	6
8	4	7	3

Output
feature map

Convolution Layer

Padding

- Input data의 주변을 특정 값으로 채움 (보통 0을 사용)
- Convolution 연산으로 boundary의 정보가 소실되는 문제점을 방지

0	0	0	0	0	0
0	1	2	3	0	0
0	0	1	2	3	0
0	3	0	1	2	0
0	2	3	0	1	0
0	0	0	0	0	0

Input Data
raw size - 4×4
after padding - 6×6



2	0	1
0	1	2
1	0	2

Filter
 3×3



7	12	10	2
4	15	16	10
10	6	15	6
8	10	4	3

Output Data
 4×4

Convolutional Neural Networks - Convolution 연산

▪ Convolution 연산의 출력 크기 계산

- 출력 크기는 filter size, stride, padding size에 따라 달라짐
- 출력 크기는 정수로 나누어 떨어져야 함

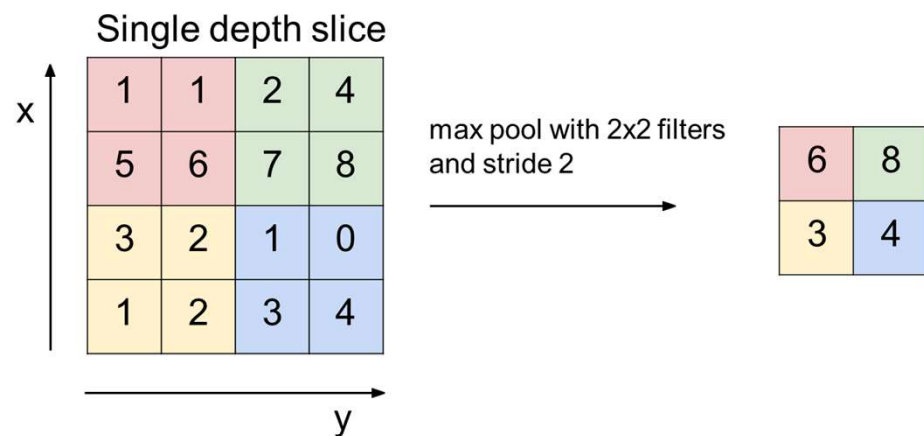
$$\text{Output Height} = OH = \frac{H + 2P - FH}{S} + 1$$

$$\text{Output Width} = OW = \frac{W + 2P - FW}{S} + 1$$

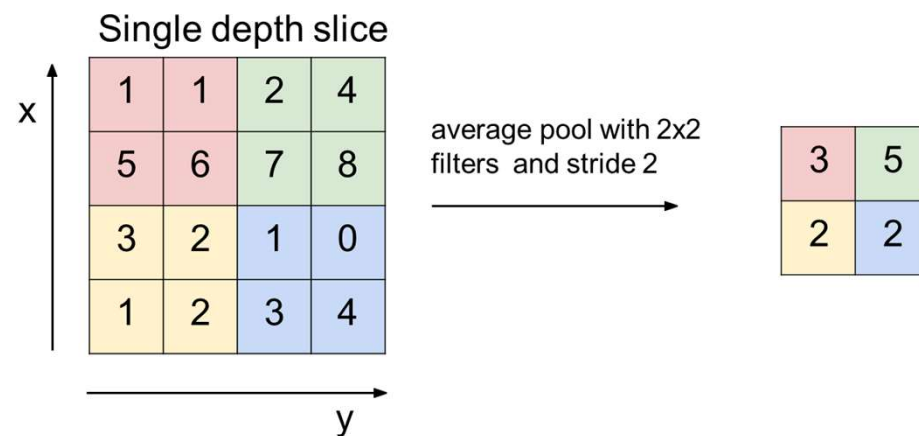
- (H, W): Input data size
- (FW, FH): Filter size
- P: Padding size
- S: Stride

Convolutional Neural Networks - Convolution 연산

- Pooling: Input data의 dimension을 subsampling 하는 방법



2x2 max pooling 예시



2x2 average pooling 예시

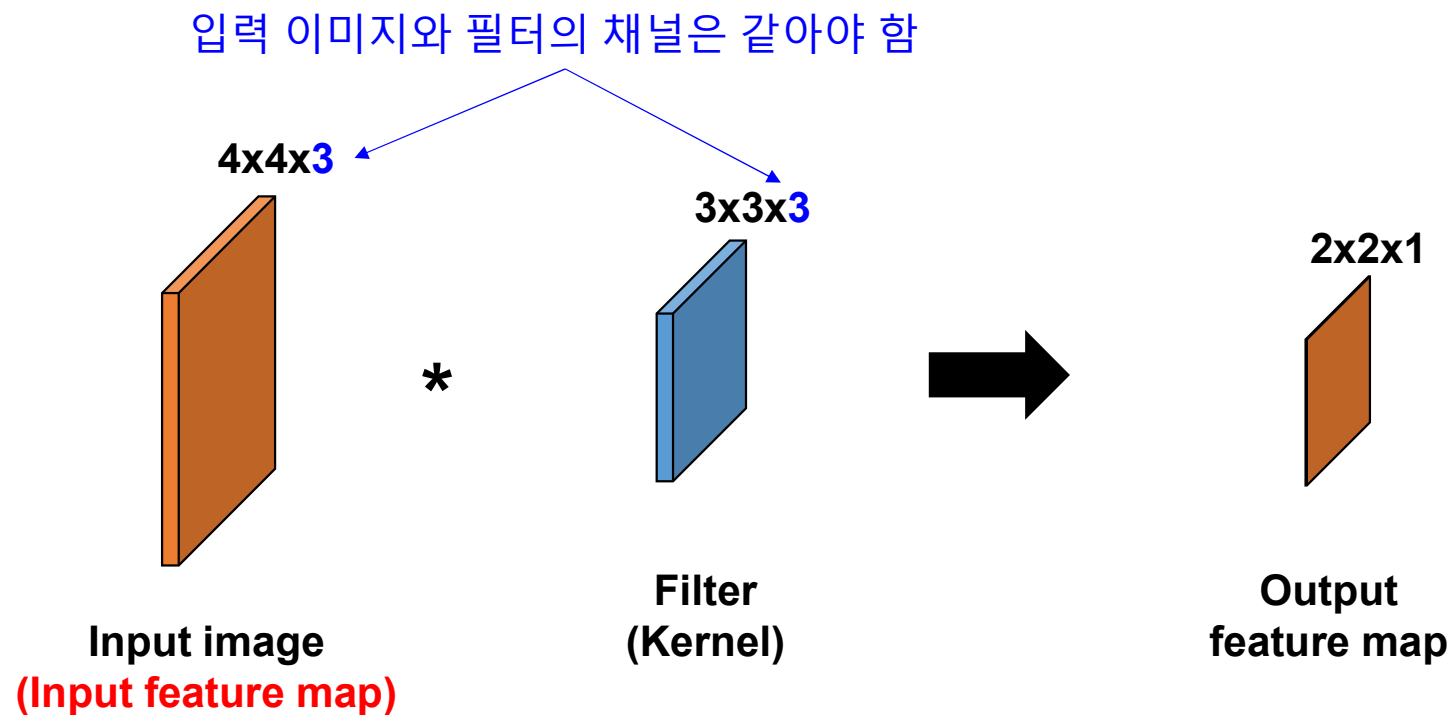
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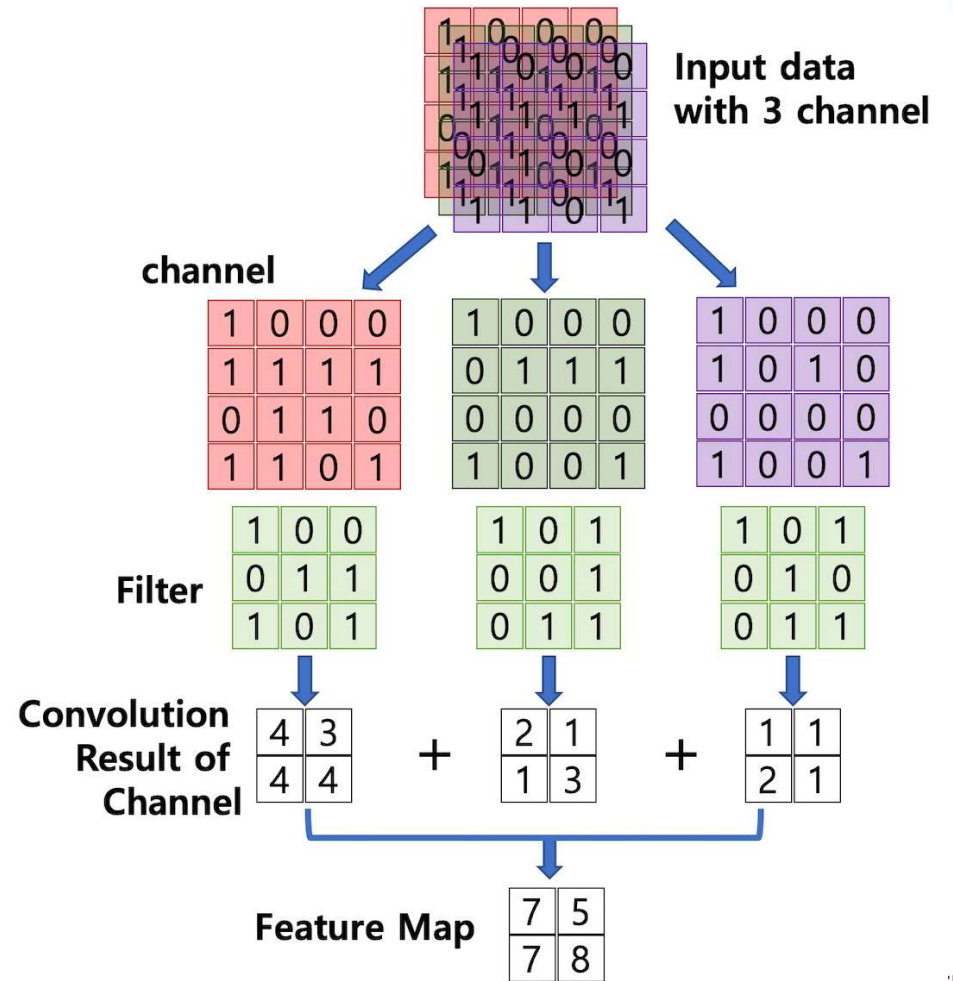
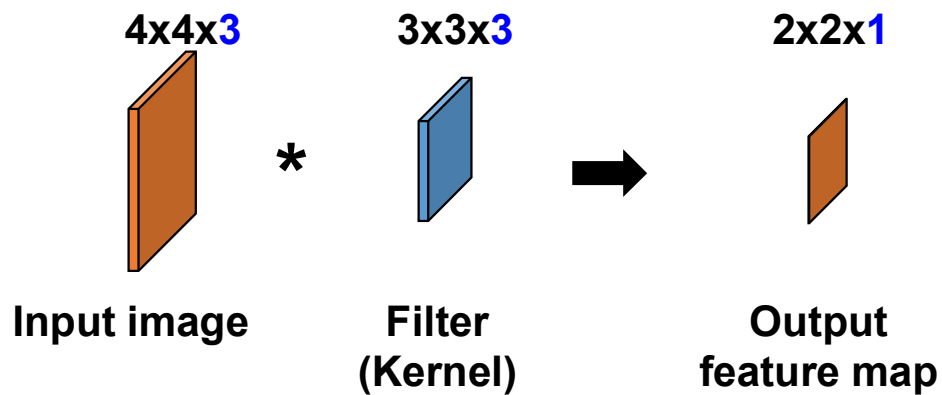
Convolutional Neural Networks - 3D 데이터의 Convolution 연산

- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산

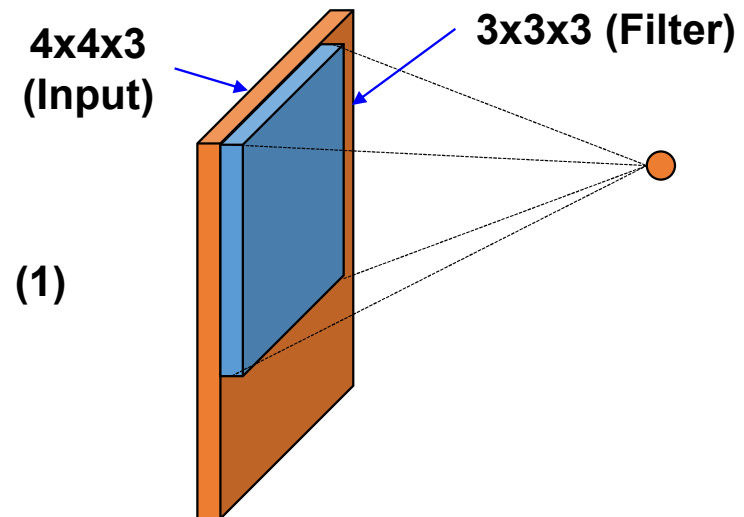


Convolutional Neural Networks - 3D 데이터의 Convolution 연산

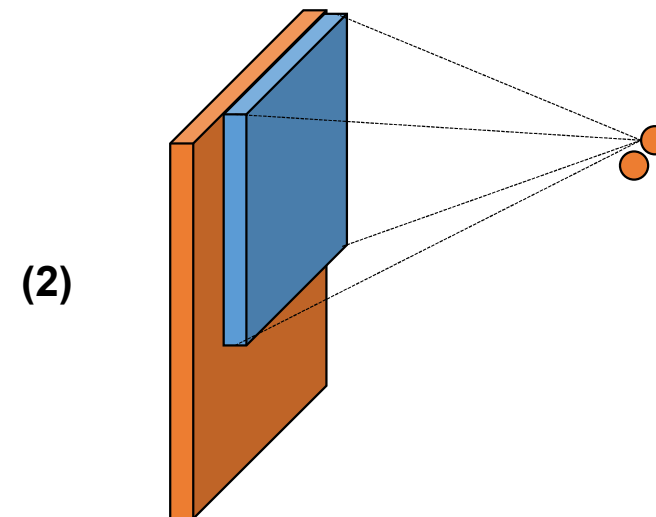
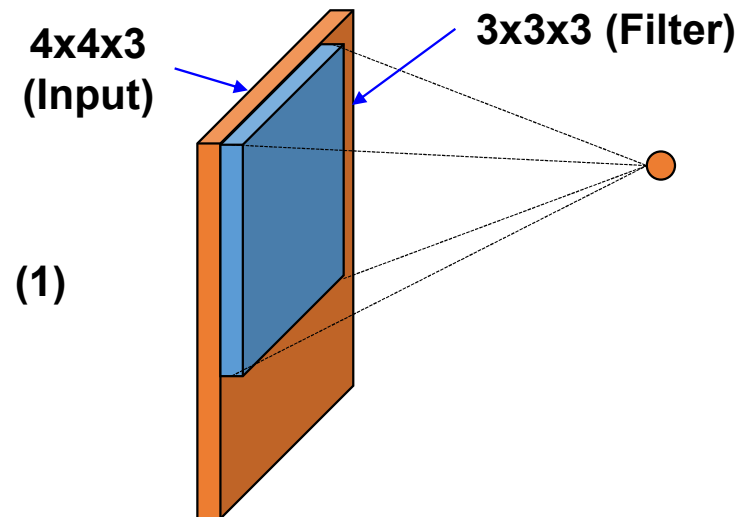
- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산



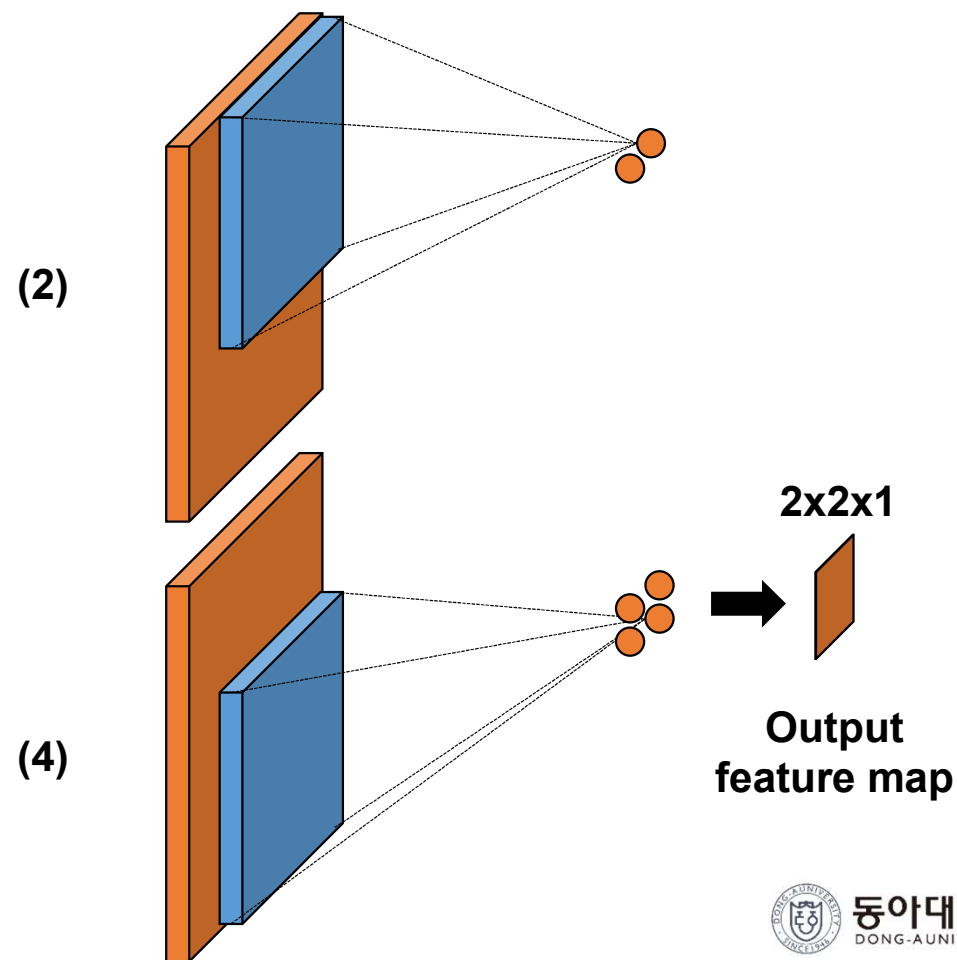
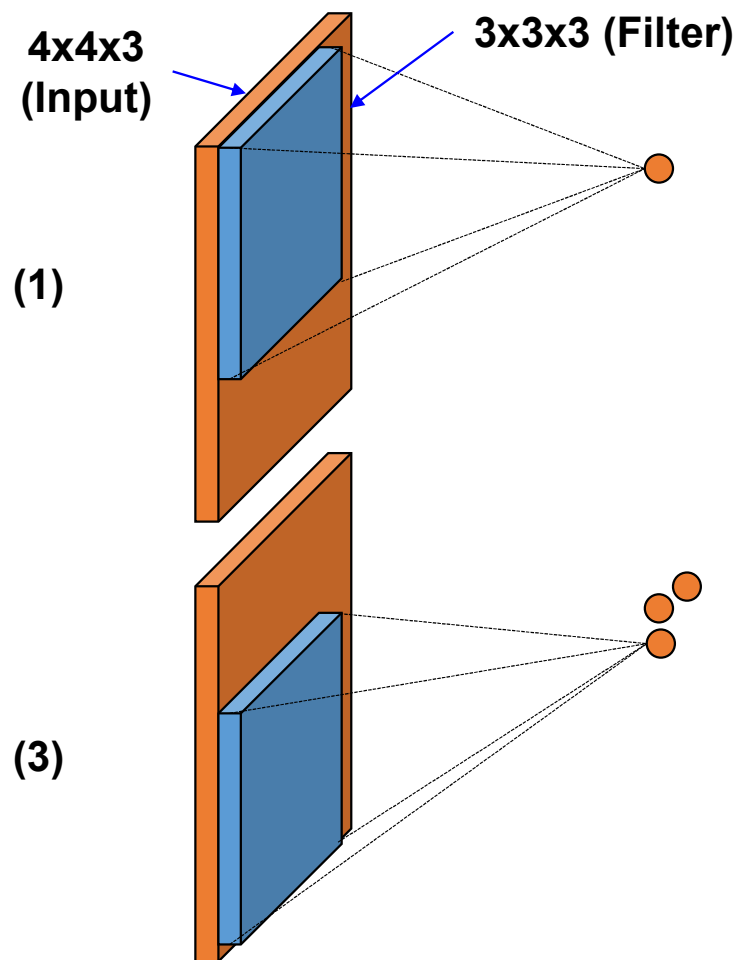
Convolutional Neural Networks - 3D 데이터의 Convolution 연산



Convolutional Neural Networks - 3D 데이터의 Convolution 연산



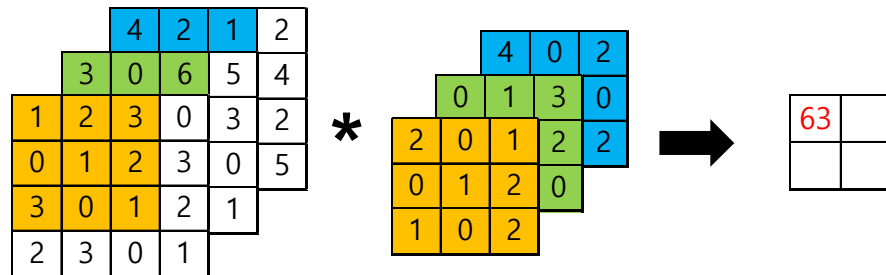
Convolutional Neural Networks - 3D 데이터의 Convolution 연산



Convolutional Neural Networks - 3D 데이터의 Convolution 연산

- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산

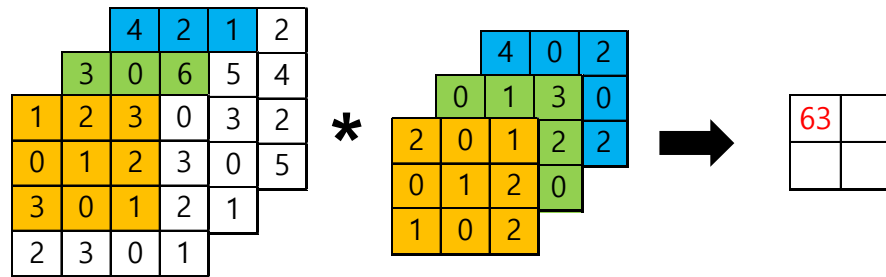
(1)



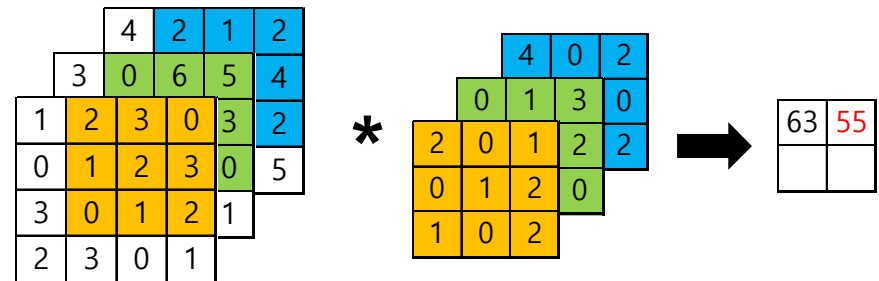
Convolutional Neural Networks - 3D 데이터의 Convolution 연산

- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산

(1)

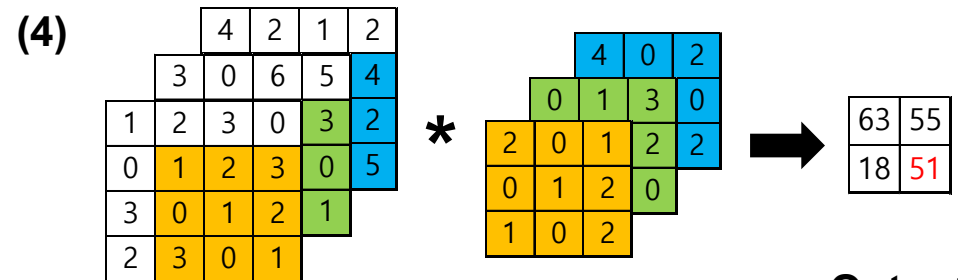
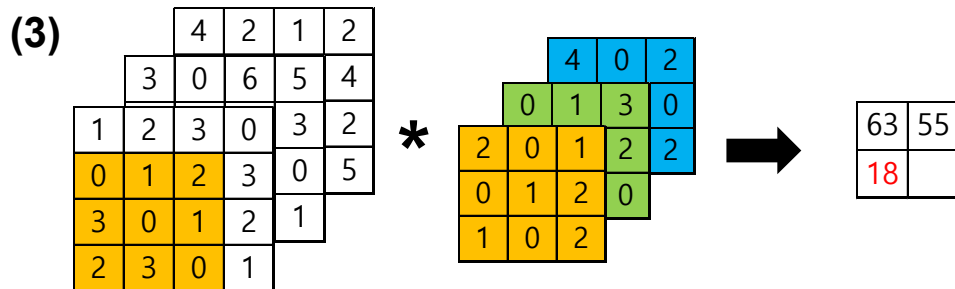
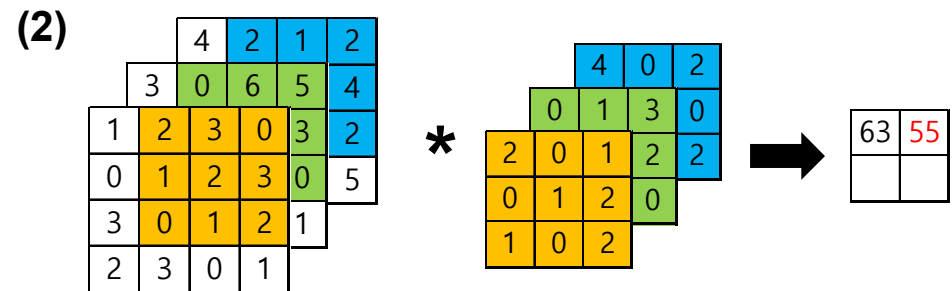
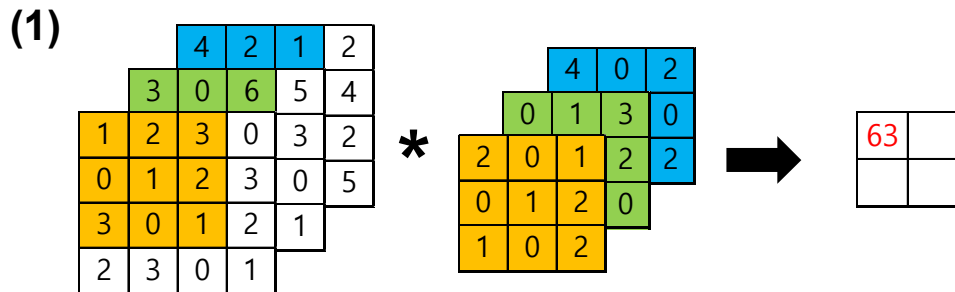


(2)



Convolutional Neural Networks - 3D 데이터의 Convolution 연산

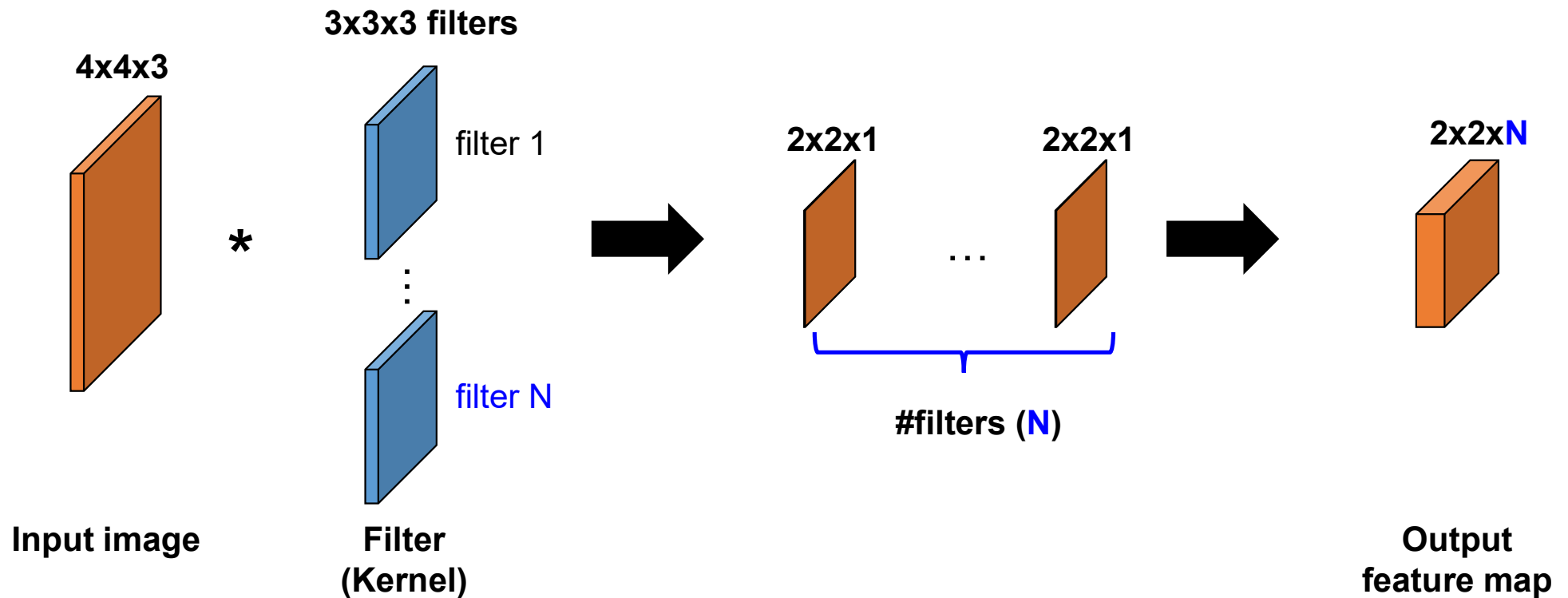
- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산



Output
feature map

Convolutional Neural Networks - 3D 데이터의 Convolution 연산

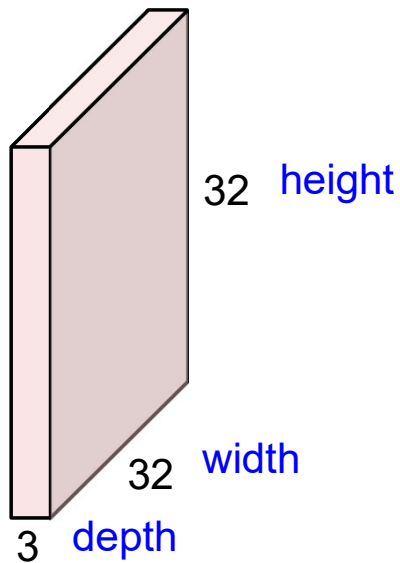
- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산
 - Output feature map의 채널 수는 filter의 개수(N)와 같음



Convolution Layer

- Overview

32x32x3 image -> preserve spatial structure



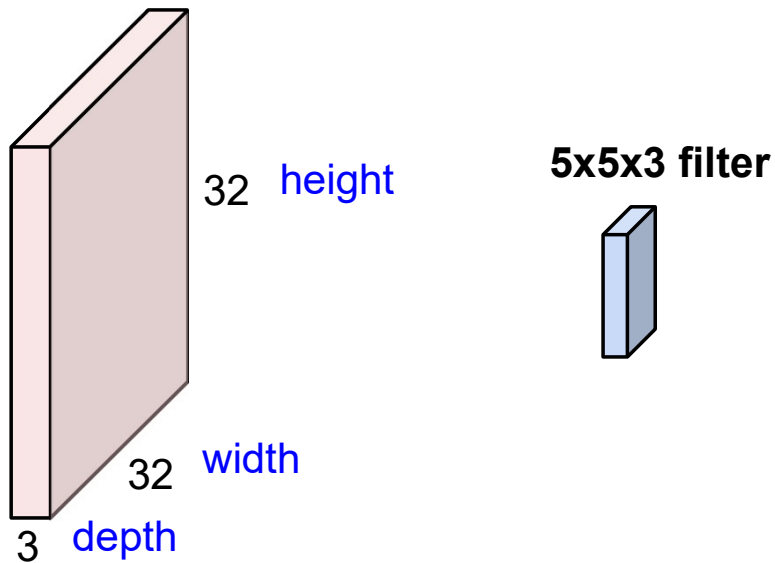
Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

32x32x3 image -> preserve spatial structure

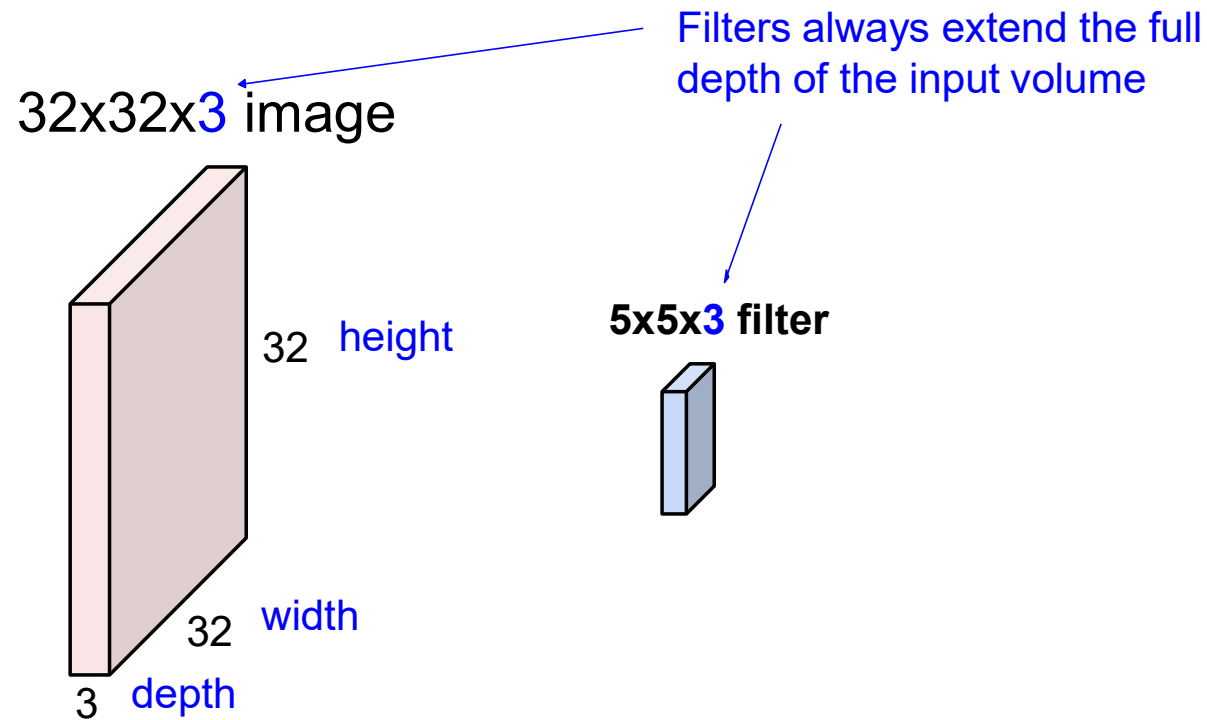


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

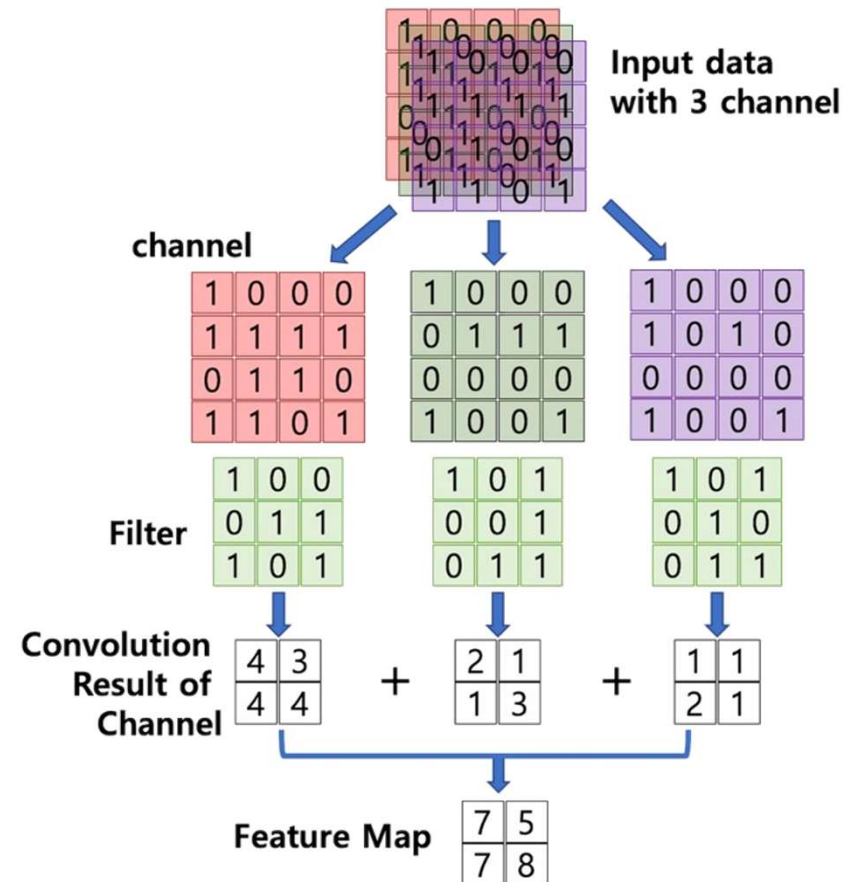
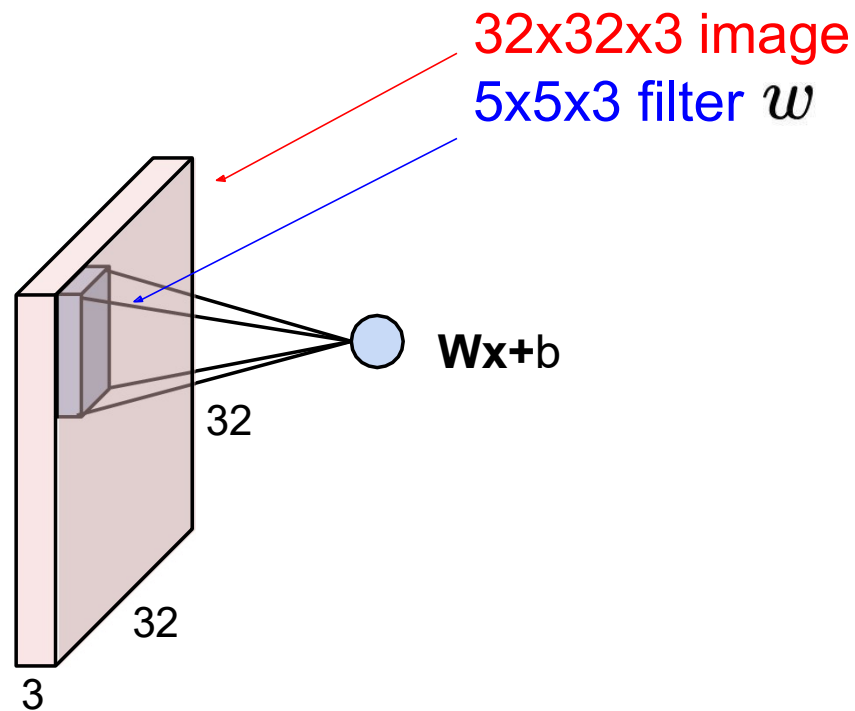


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

Overview

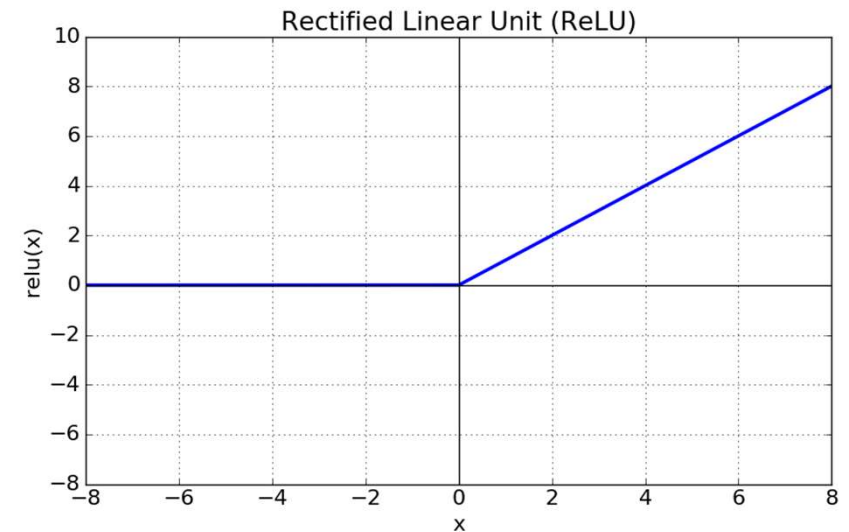
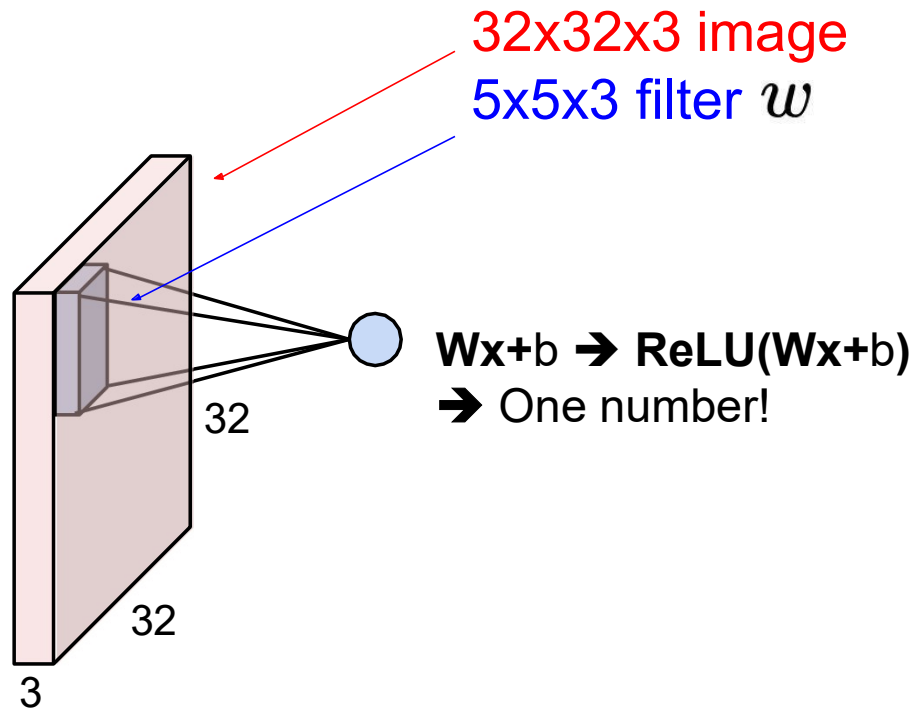


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

Overview

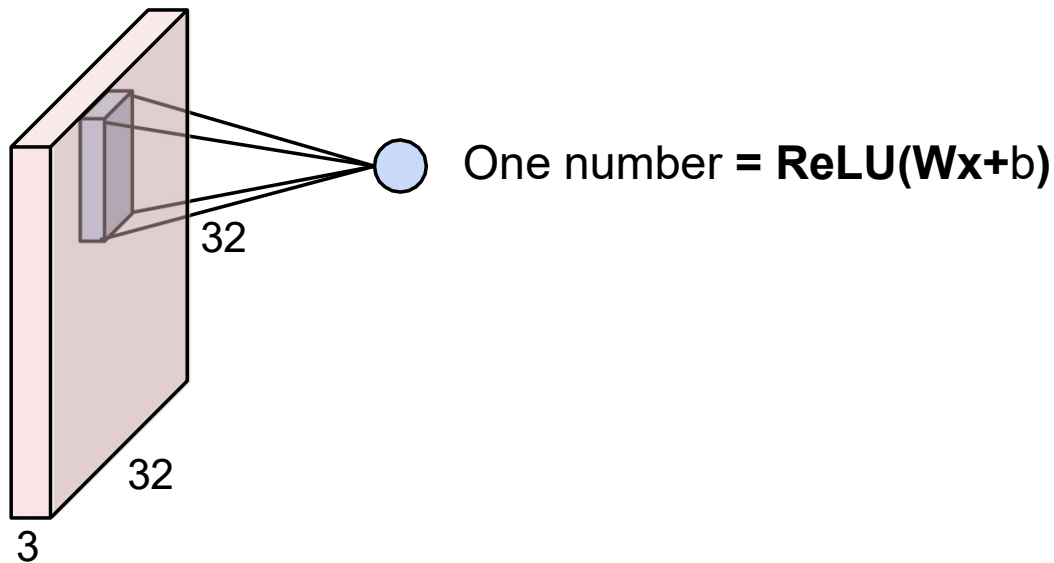


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

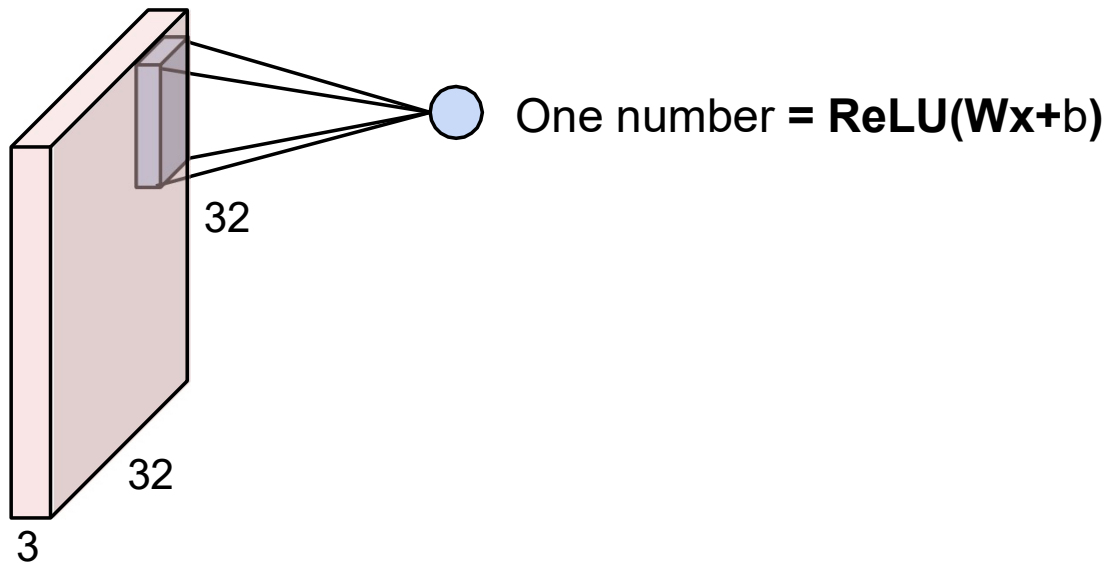


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

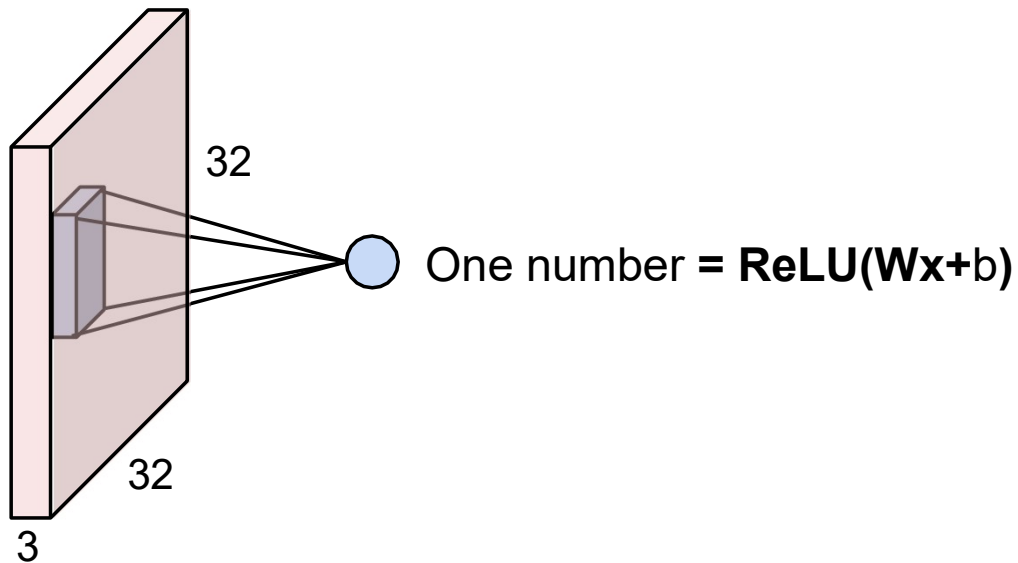


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

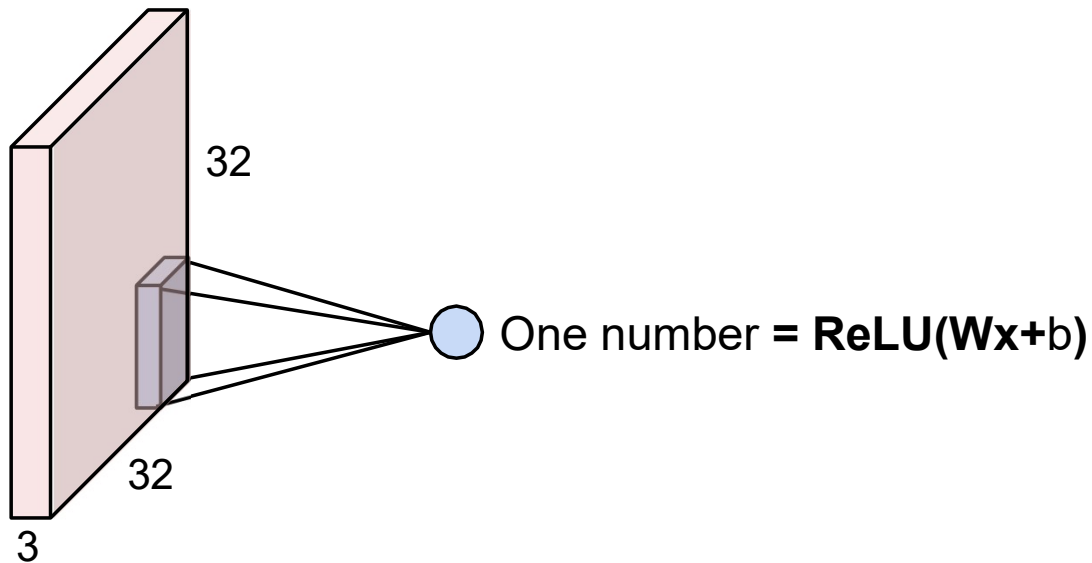


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

- Overview

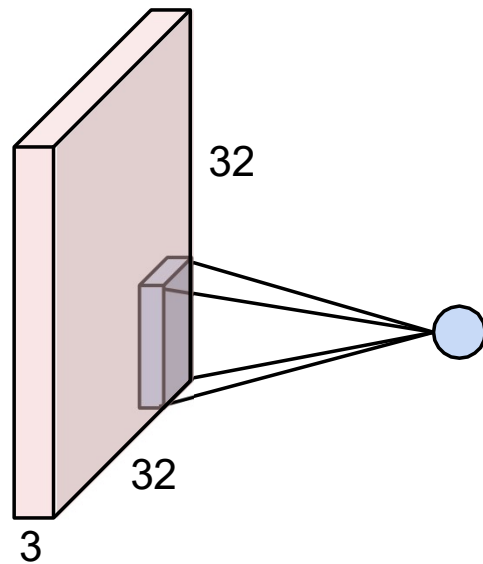


Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

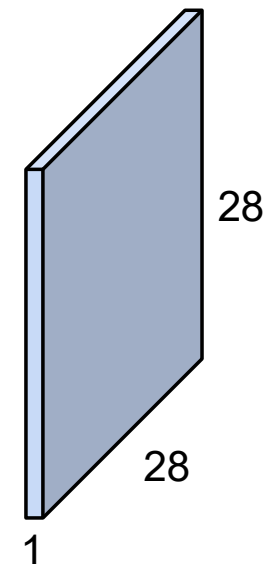
Convolution Layer

- Overview



convolve (slide) over all spatial locations

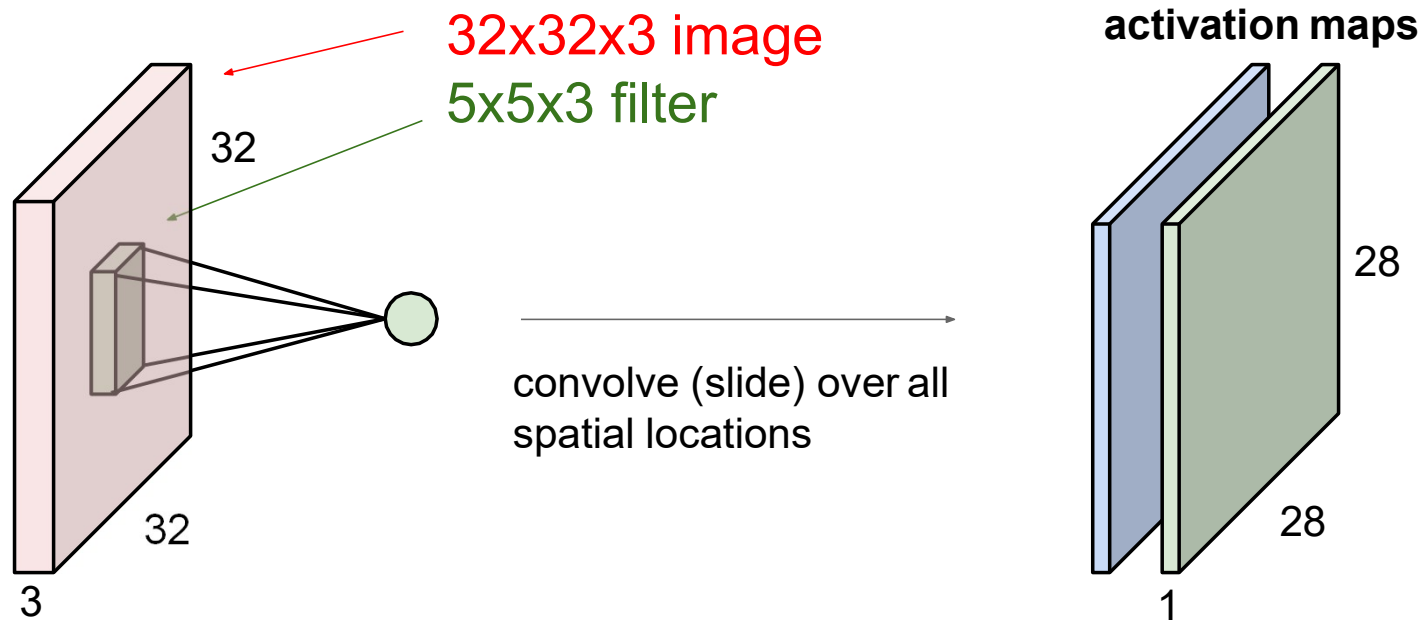
activation map



Convolution Layer

Overview

- Consider a second, green filter



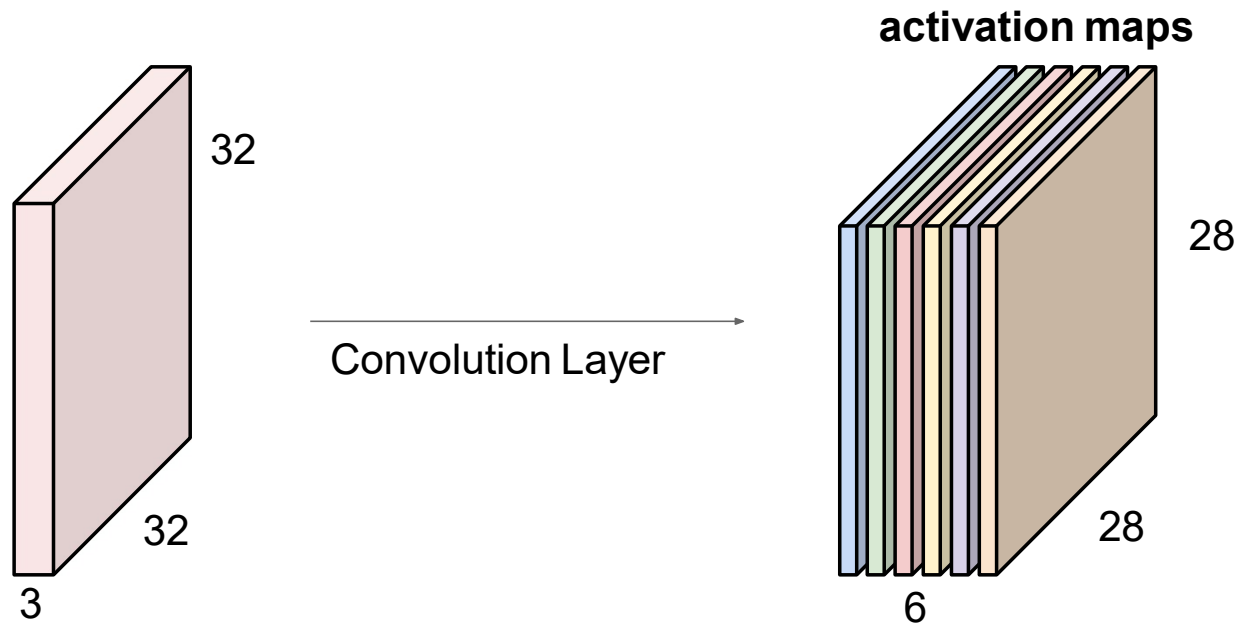
Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

Overview

- For example, if we had 6 $5 \times 5 \times 3$ filters, we'll get 6 separate activation maps



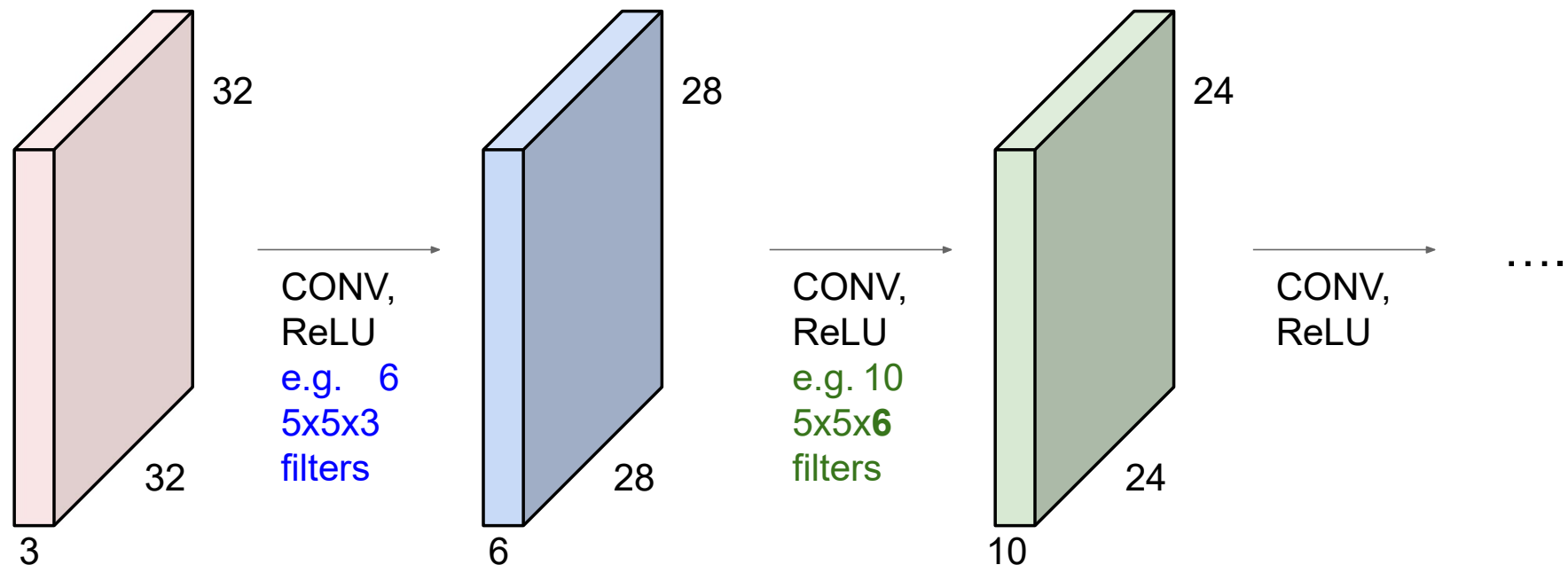
Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolution Layer

Overview

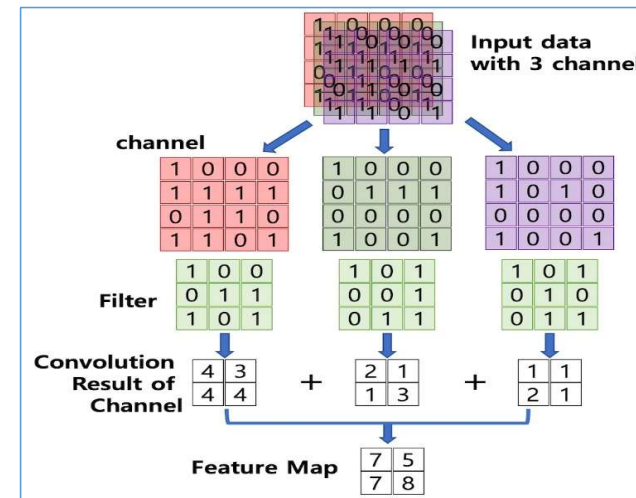
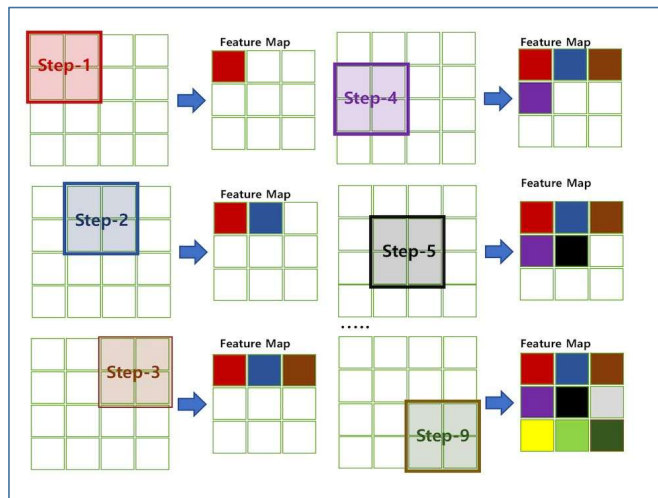
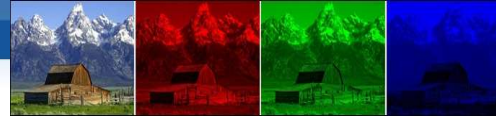
- ConvNet is a sequence of Convolutional Layers, interspersed with activation functions



Fei-Fei Li & Justin Johnson & Serena Yeung

출처: cs231n_2017_lecture5 April 18, 2017

Convolutional Operation



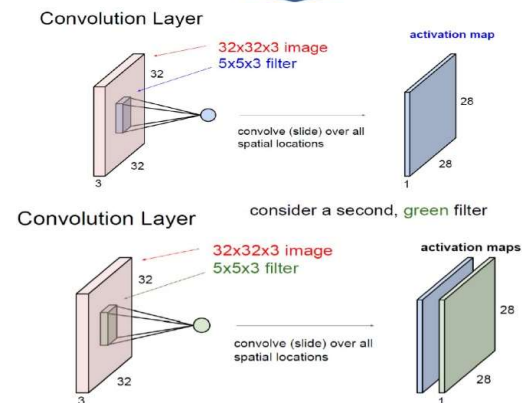
Convolution

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Image

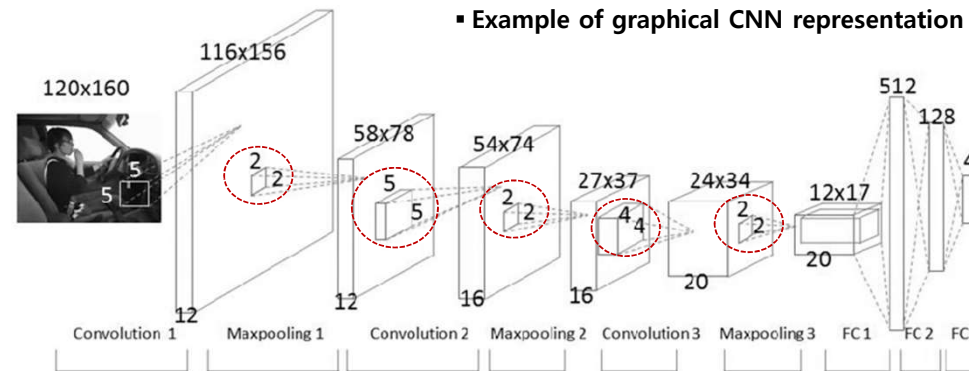
4		

Convolved Feature

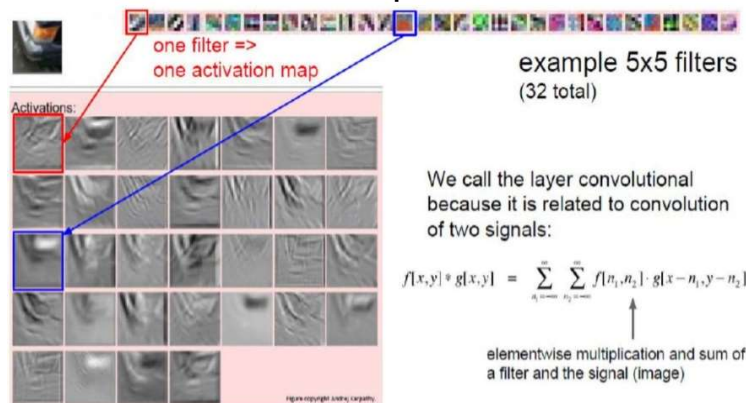


Deep Learning: CNN Visualization

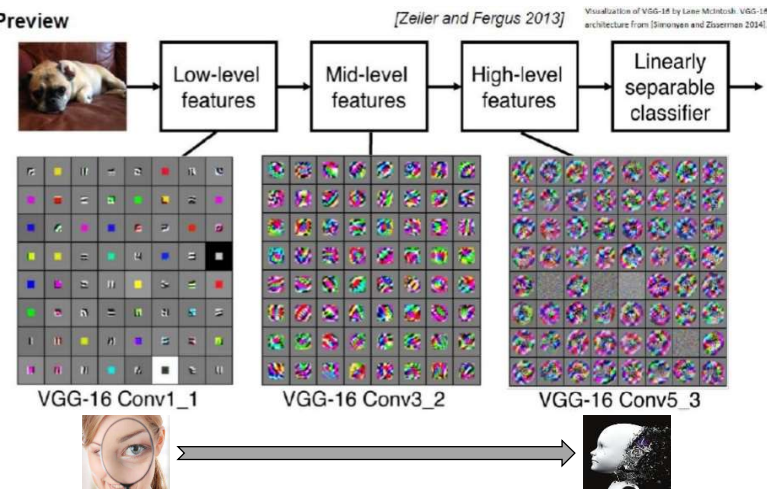
▪ Example of graphical CNN representation



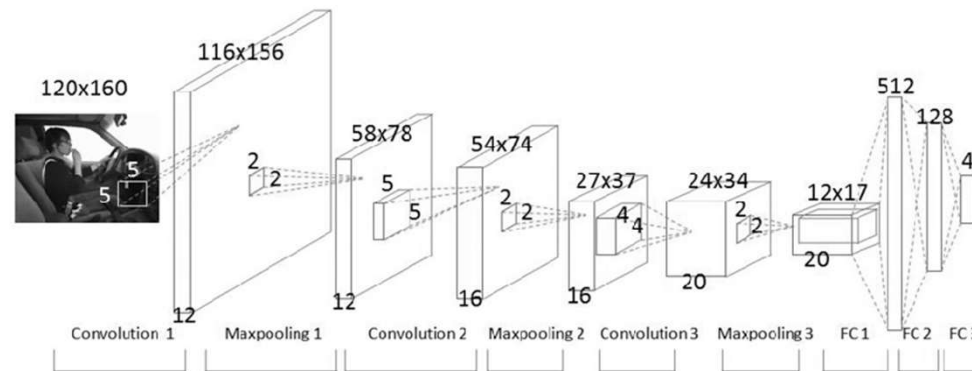
▪ CNN trains the multiple filters (Kernel).



Preview



CNN Implementation Keras

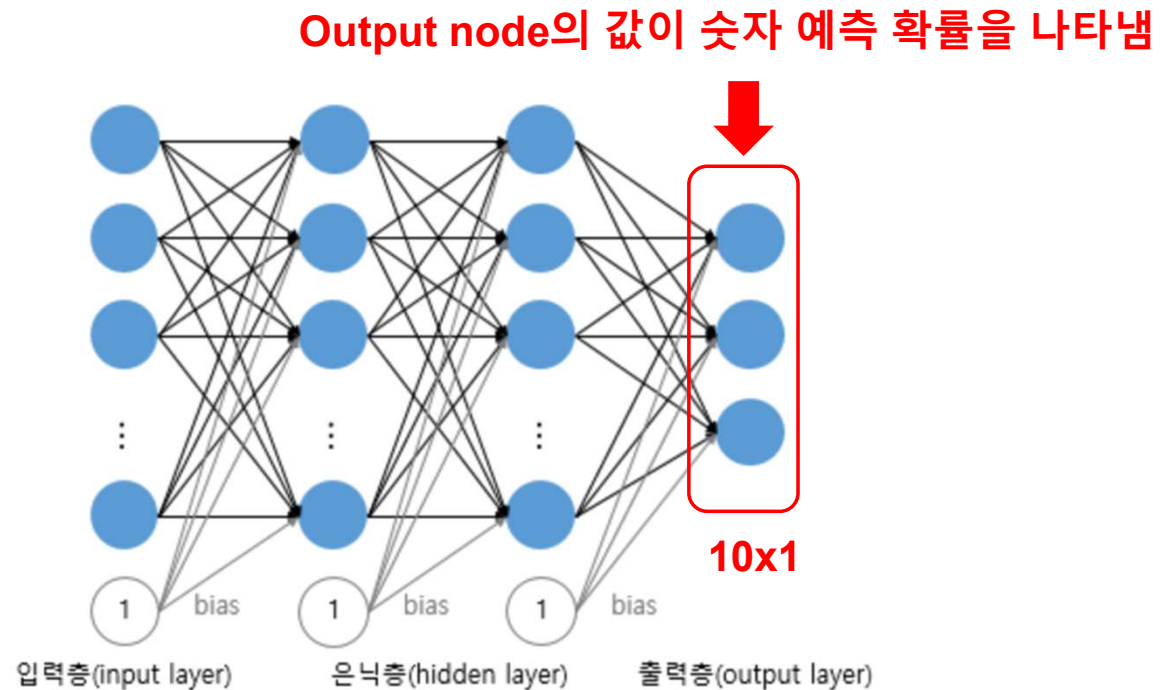
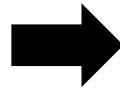
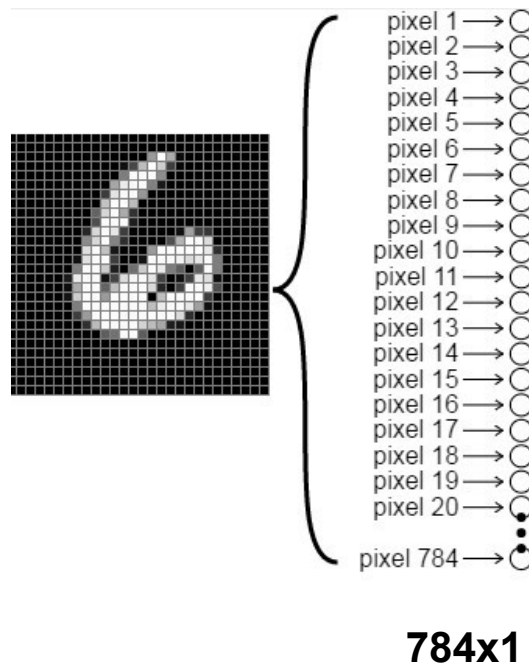


```
from keras.models import Sequential
from keras.layers.convolutional import Conv2D
from keras.layers.convolutional import MaxPooling2D
from keras.layers import Dense
from keras.layers import Flatten

model = Sequential()
model.add(Conv2D(12, kernel_size=(5, 5), activation='relu', input_shape=(120, 60, 1)))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Conv2D(16, kernel_size=(5, 5), activation='relu'))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Conv2D(20, kernel_size=(4, 4), activation='relu'))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Flatten())
model.add(Dense(128, activation='relu'))
model.add(Dense(4, activation='softmax'))
```

Convolutional Neural Network (CNN) 이론

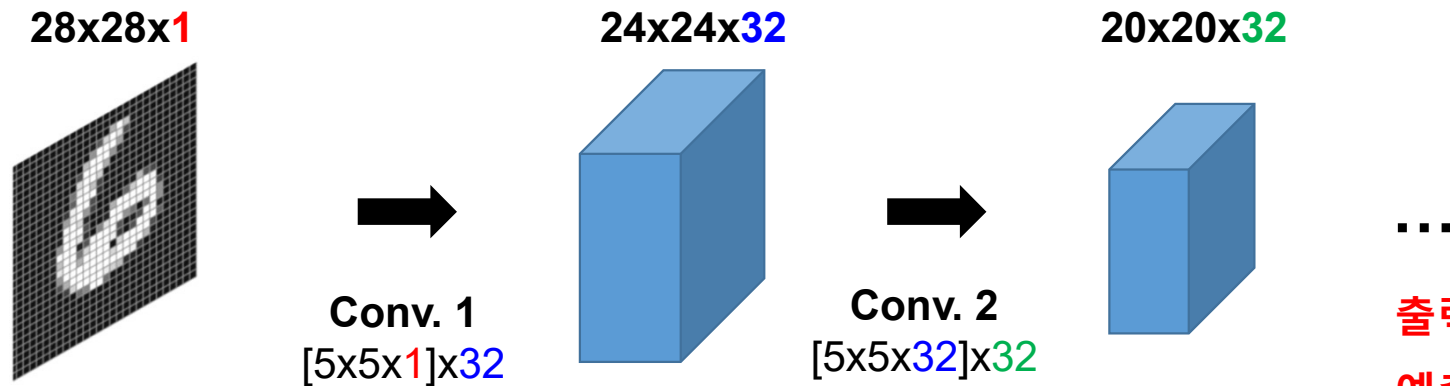
- CNN을 이용한 classification model 설계 시 주의사항



MLP model 예시

Convolutional Neural Network (CNN) 이론

- CNN을 이용한 classification model 설계 시 주의사항

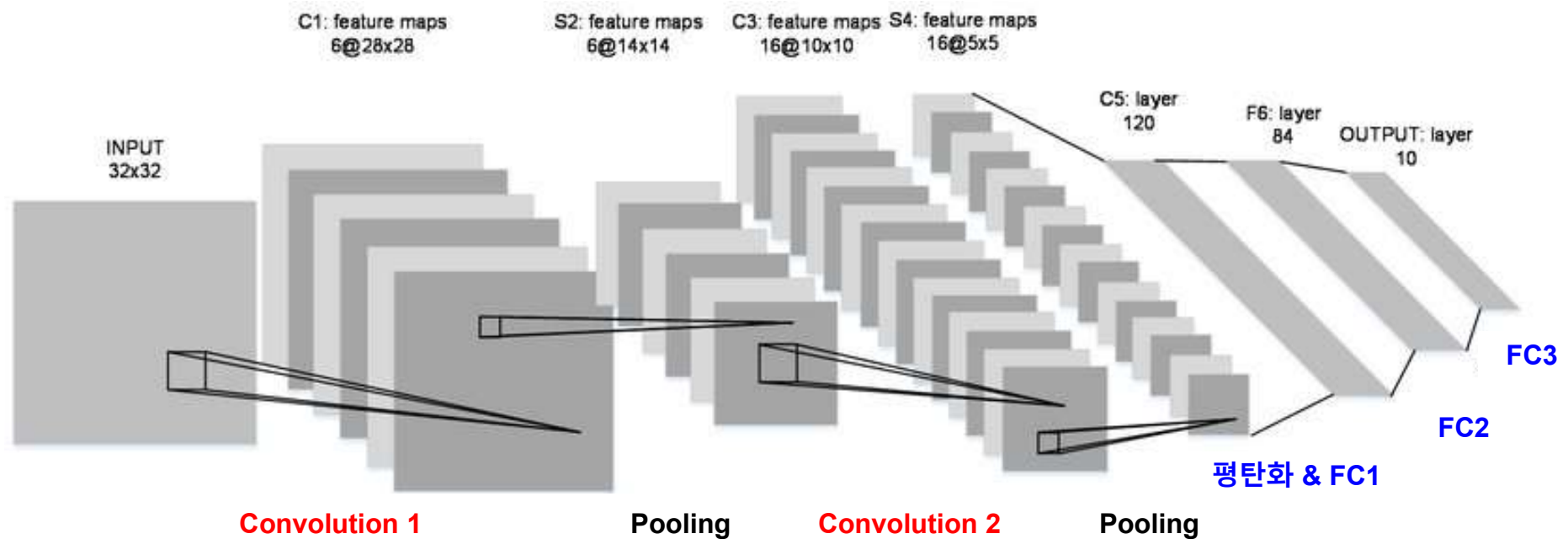


...
출력의 형태가 cube 이므로
예측 확률을 확인 할 수 없음

CNN model 예시

Convolutional Neural Network (CNN) 이론

- CNN을 이용한 classification model 설계 시 주의사항
 - 일반적으로 CNN의 feature map을 평탄화 한 이후 fully connected layer에 입력함



LeNet-5 구조

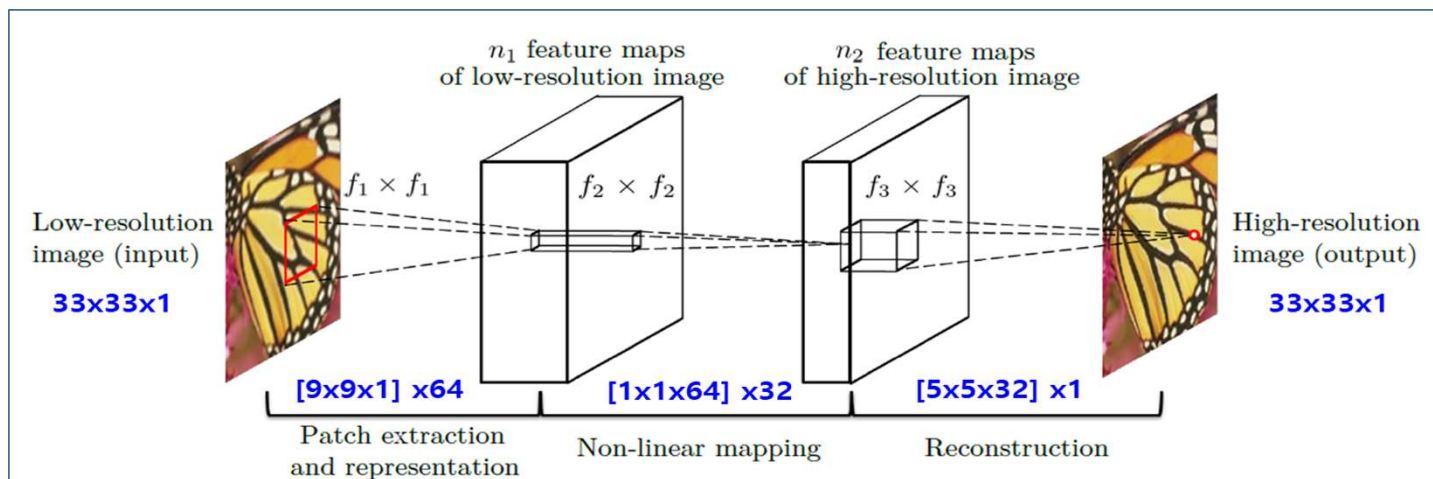
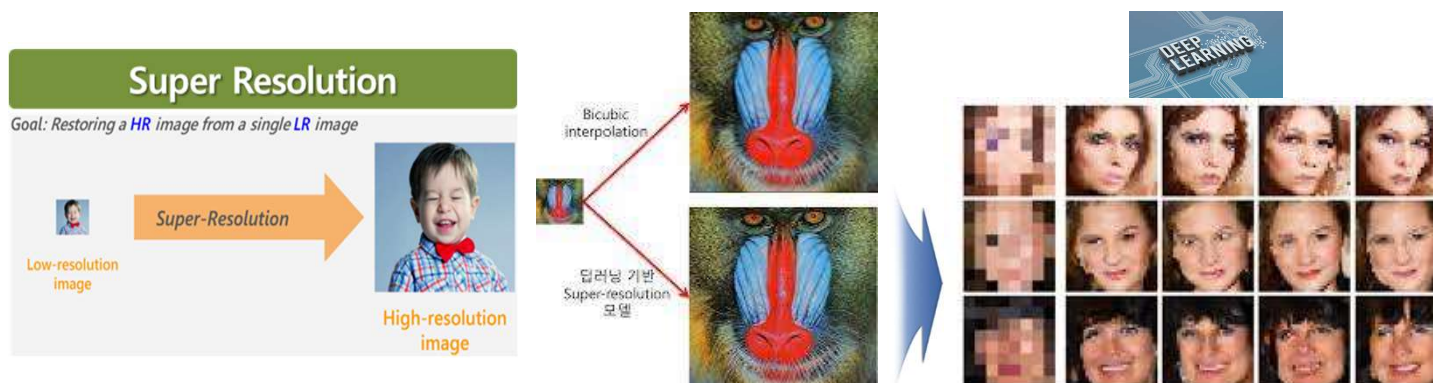
LeNet-5 구조

- C1** 훈련해야할 파라미터 개수: $(가중치 * 입력맵개수 + 바이어스) * 특성맵개수 = (5 * 5 * 1 + 1) * 6 = 156$
- S2** 훈련해야할 파라미터 개수: $(가중치 + 바이어스) * 특성맵개수 = (1 + 1) * 6 = 12$
- S4** 훈련해야할 파라미터 개수: $(가중치 + 바이어스) * 특성맵개수 = (1 + 1) * 16 = 32$
- C5** 훈련해야할 파라미터 개수: $(가중치 * 입력맵개수 + 바이어스) * 특성맵 개수 = (5 * 5 * 16 + 1) * 120 = 48120$
- F6** 훈련해야할 파라미터 개수: $연결개수 = (입력개수 + 바이어스) * 출력개수 = (120 + 1) * 84 = 10164$

Applications

Applications

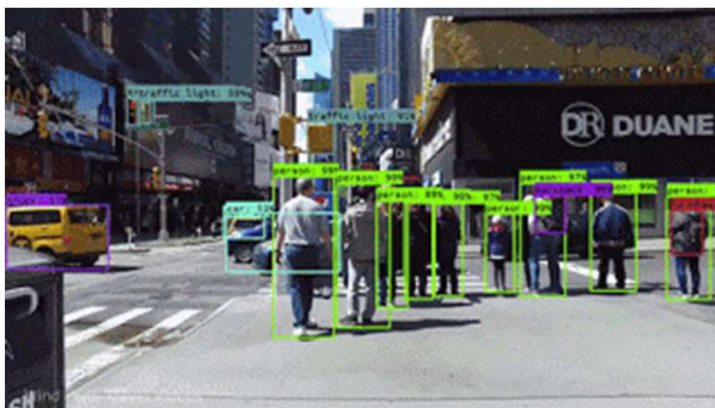
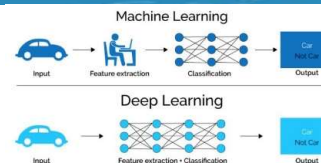
CNN based Super Resolution (SR)



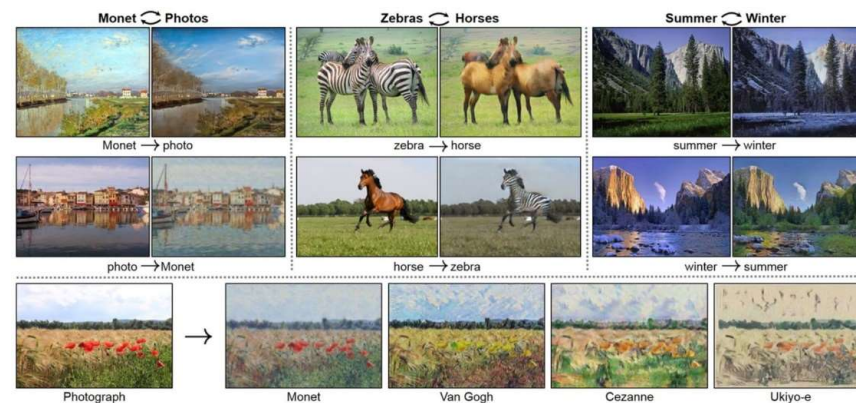
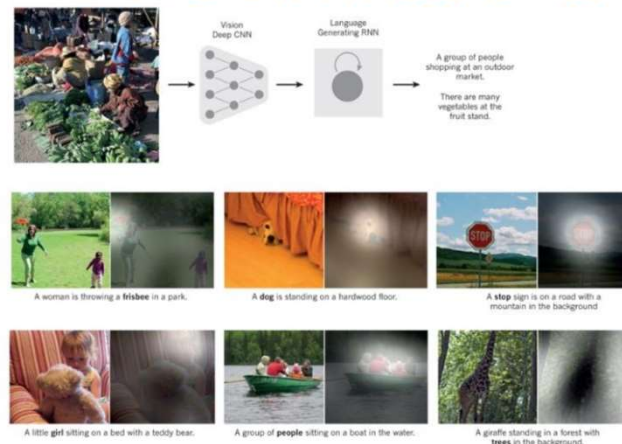
* Reference: IEEE Trans. PAMI "Image Super-Resolution Using Deep Convolutional Networks" Chao Dong et al, 2015

Applications

Paradigm Shift (Image/Video Processing)



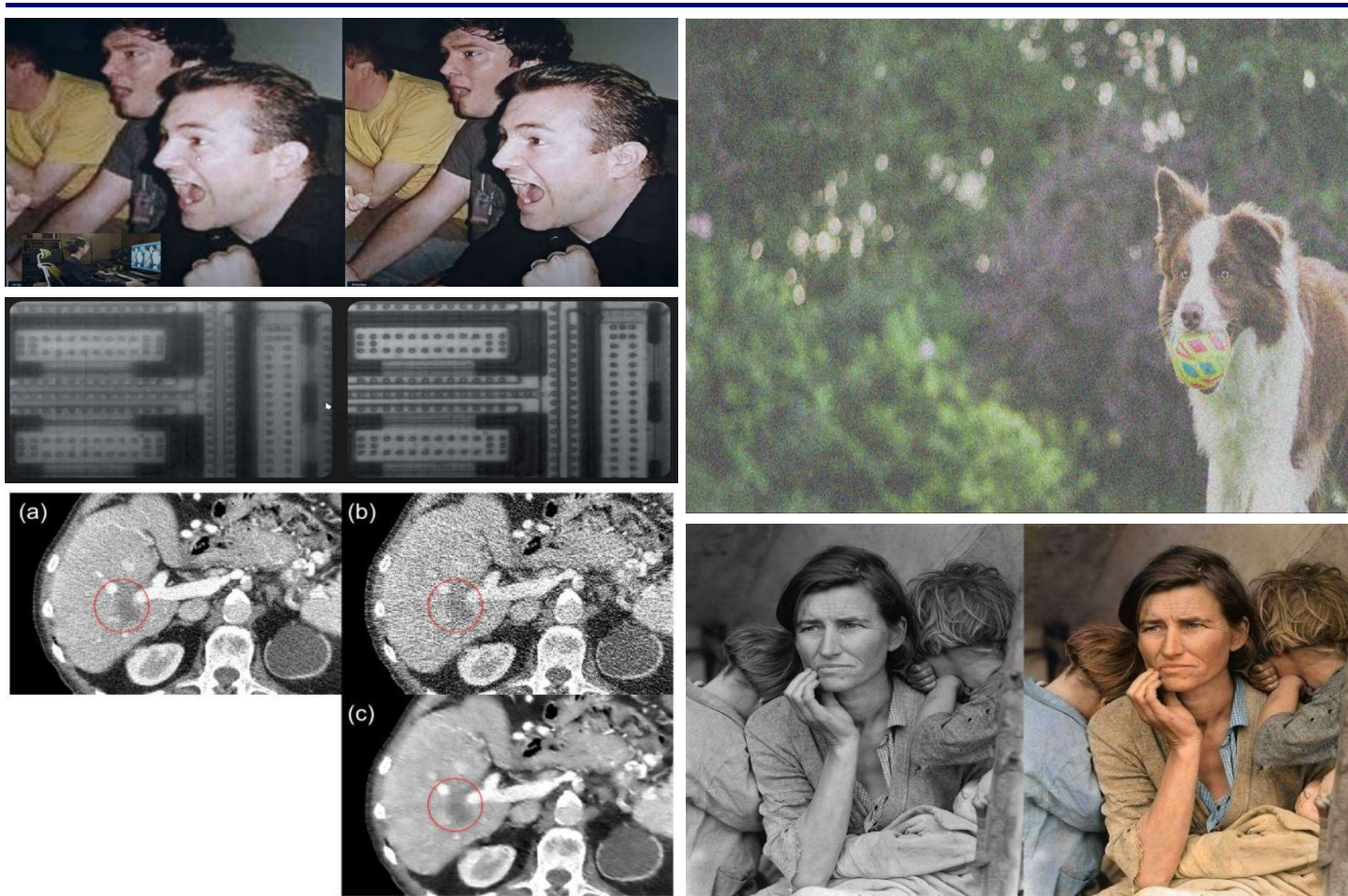
CNN과 RNN을 조합하여 이미지의 캡션을 자동으로 생성



CycleGAN Project Page (<https://junyanz.github.io/CycleGAN/>)

Applications

SR Applications (Image Restoration)



Applications

SR Applications (Video Restoration)

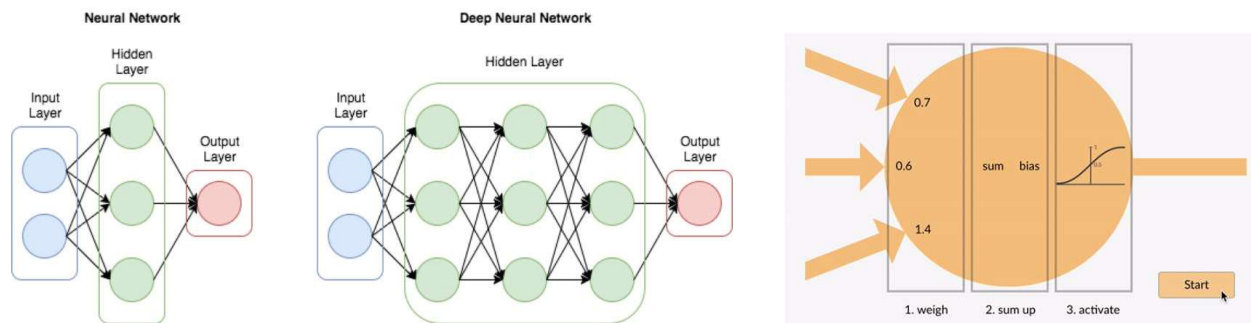
Source



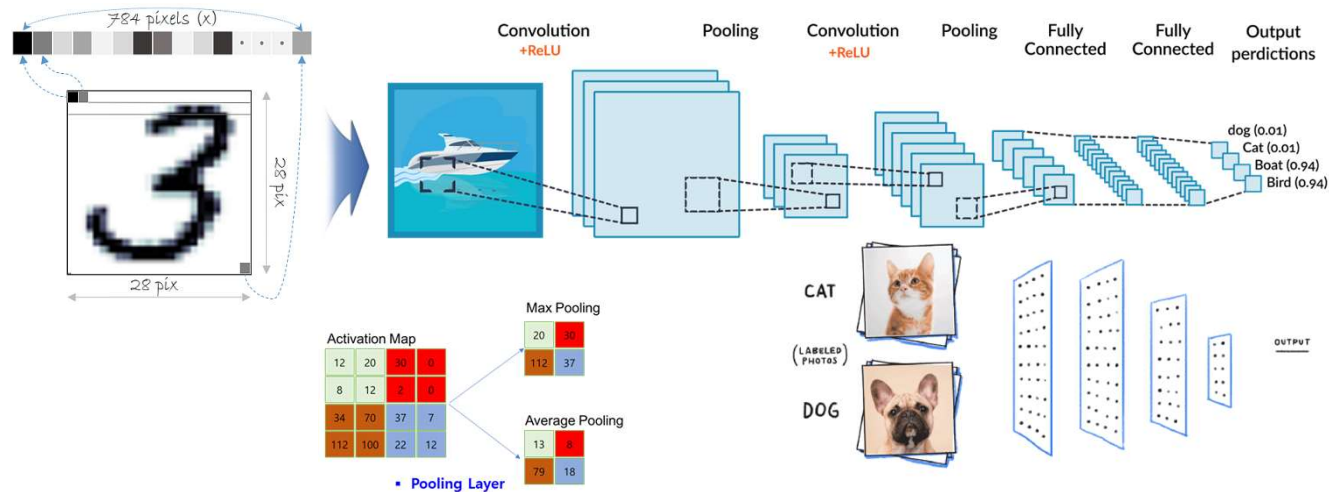
Conclusions

MLP vs CNN

Artificial Neural Network (ANN) → Deep Neural Network (DNN)

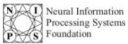


Convolution Neural Network (CNN)



References

The world's top 5 conferences

Rank		Conference Details	H5-index	Impact Score
1		IEEE/CVF Conference on Computer Vision and Pattern Recognition 2022-06-21 - 2022-06-24 - New Orleans (CVPR)	299	51.98
2		Neural Information Processing Systems (NIPS) 2021-12-06 - 2021-12-14 - Online (NIPS)	198	33.49
3		IEEE/CVF International Conference on Computer Vision 2021-10-11 - 2021-10-17 - Montreal (ICCV)	176	32.51
4		European Conference on Computer Vision 2021-10-11 - 2021-10-17 - Montreal (ECCV)	144	25.91
5		AAAI Conference on Artificial Intelligence 2022-02-22 - 2022-03-01 - Vancouver (AAAI)	126	25.57



19.JUNE VIRTUAL, US

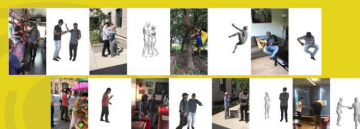
NTIRE 2021

New Trends in Image Restoration and Enhancement workshop
and challenges on image and video processing
in conjunction with CVPR 2021

ECCV'20
SEC, Glasgow
23-28 AUGUST 2020

FIRST 3D POSES IN THE WILD CHALLENGE

SEC, GLASGOW
23 AUGUST 2020





Mobile AI Workshop 2021

Designing the Next-Generation AI Models for Mobile Devices...

20 June, 7:00 PT Virtual, US CVPR 2021

16.OCTOBER MONTREAL, CANADA

AIM 2021

Advances in Image Manipulation workshop
in conjunction with ICCV 2021

LIVE ON ZOOM IMPORTANT DATES NTIRE2021 MAT2021 AIM2021



Questions & Answers

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